

TECHNICAL UNIVERSITY OF CLUJ-NAPOCA

DEPARTMENT OF DESIGN ENGINEERING AND ROBOTICS

MASTER'S DEGREE IN ROBOTICS

RESEARCH PROJECT

STUDENT

CĂLIAN ALEXANDRU - ȘTEFAN

SERIES OF STUDY 2023-2025

**TECHNICAL UNIVERSITY OF CLUJ-NAPOCA
DEPARTMENT OF DESIGN ENGINEERING AND ROBOTICS**

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CĂLIAN ALEXANDRU - ȘTEFAN

**DESIGN AND IMPLEMENTATION OF A ROBOTIC SYSTEM FOR HIGH
PRECISION GLASS FACADE CLEANING**

SCIENTIFIC COORDINATOR

Prof. Dr. Ing. Stelian Brad

1ST SEMESTER

PROJECT PRESENTATION

1. VISION

This project has in mind the design and implementation of a robotic system for high precision glass façade cleaning. It will mainly consist of the initial documentation phase where I will study the working principle of said system while also consulting the current stage of development.

Following this, I want to study and take into consideration every component that is involved, from the mechanical structure, to the technologies used and the train of thought in utilizing the most effective and efficient software. After this I will need to do the required calculation with regards to the physical mechanism of the robot and also the programming part of it. A physical or simulated design of the robotic system will need to also be created so that in the final phase I can readjust and correct existing problems and eventually improve the existing solution.

2. OBJECTIVES AND ACTIVITIES PLANNED FOR SEMESTER 1

For the first semester I plan to accustom myself with the current subject, study the current solutions that have been or are currently in development, and try to create a personal solution that incorporates the best solutions that have been considered or approached in the already existing ones.

3. OBJECTIVES AND ACTIVITIES PLANNED FOR SEMESTER 2

For the second semester I want to start doing the calculations necessary in creating the physical mechanism and structure of the robotic system while also taking into consideration the software and programming solutions that need to be developed.

4. OBJECTIVES AND ACTIVITIES PLANNED FOR SEMESTER 3

For the third semester I plan on recreating the robot in a physical representation or in a simulated environment.

5. OBJECTIVES AND ACTIVITIES PLANNED FOR SEMESTER 4

For the last semester I want to revisit the work realised on the course of the previous semesters in hope that i can readjust or fix certain errors or even try to improve the current stage of the project.

DOCUMENTATION ON THE RESEARCH TOPIC

1. Introduction in the project theme

Before diving into the project i would like to make a introduction to the current state of development regarding this theme while also emphasising the importance of said developments in current industry, and in the near future.

In our current time period glass façade cleaning is very important in maintaining a building simply because of the environment elements that degrade the glass over time if not cleaned diligently. Dirt, sand and bird residue are the most common elements that stain the glass of the façades deteriorating it over time which is the most important hazard when we talk about glass maintenance. Periodical cleaning is also important in keeping up the appearance of a building, especially if we are talking about a skyscraper or more likely a company building.

We need to take into consideration that a good part of the current population is linking the way a building is presented itself, to the direct representation of the diligence and responsibility of the people that maintain it.

When we talk about glass façade cleaning, we talk about two different solutions:

- manual labor cleaning
- semi-automatic/fully automated cleaning

1.1. Manual labor cleaning

Manual glass façade cleaning is the most common one out of the two, simply because recruiting more human working force is easier and less costly than actually developing a robotic system for it, not to forget about the maintaining cost that is even cheaper than maintaining the robot.

These are the most important advantages that “classic” façade cleaning brings, but also the only ones because outside of these we got only disadvantages, especially when talking about the risks that need to be taken into the equation.



Figure 1.1 Rope scaffolding facade cleaning



Figure 1.2 Scaffold facade glass cleaning

a) Personnel training

The first disadvantage that we need to take into consideration is the cost of training the workers because is one of the most crucial factors when trying to improve the quality of the service while also diminishing the risk of injury at the working place. A considerate amount of time will be taken into correctly instructing the workers to utilize the scaffolding tools in a safe environment, making routine checks for the maintenance of the tools, using proper techniques in ensuring their own safety while also doing their job efficiently.

b) Working at height

Cleaning high buildings and skyscrapers present a higher risk than cleaning at a smaller scale simply because of the height, which poses the most common cause of workplace injuries and fatalities in the United Kingdoms as told in an article posted on the page of the insurance company Markel. Data from Reporting of Injuries, Diseases and Dangerous Occurrences Regulations known as RIDDOR, shows that this practice was to blame for 8% of all non – fatal injuries at work during the time period of 2021-2022, while 29 workers lost their lives while falling from height in the same time period.

Even if the Work at Height Regulations that were revisited in 2005, state that window cleaners may work from height only when a different approach is not possible, most of the situations call for scaffolding the building.

c) Injuries at the working place

Because of the limited space in which a worker has to clean the façade, the risk of injury is higher than most of the working environments. Window cleaners are prone to RSI or repetitive strain injury that can develop if you repeatedly use one part of your body in a certain way, in this case repeatedly using the arms and shoulders in reaching, pulling and pushing motions. The right use of protective equipment while also taking regular breaks will lower the risk of contracting this kind of injuries.

d) Property damage

Because the job of a window cleaner brings you into contact with members of the public and their property, no matter how careful or trained you are, there is always a risk of inadvertently damaging someone's property during work.

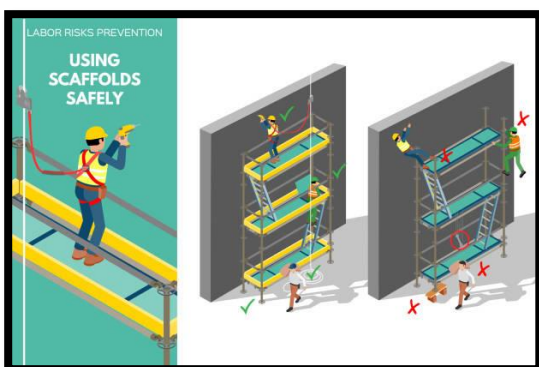


Figure 1.3 Scaffolding safety instructions



Figure 1.4 Work hazard warning

1.2. Robotized cleaning

All these risks and disadvantages presented in the previous pages, will directly influence the quality of the service, while also compromising on the efficiency regarding the time needed to complete the cleaning and the profit.

This is why the requirement for clean and safe working conditions has driven the advancement of robotic innovations that are used in façade glass cleaning.

The evident advantage is the lack of personnel involved in the cleaning of the façade glass. Depending on the cleaning solution adopted in developing the robot, there is needed only a single person that is tasked with controlling and adjusting the robot's path in the case of semi-automated robots, or none at all if we are talking about fully automated robotic systems.

Another considerable advantage is the fact that robotized solutions are capable of reaching location on the façade that were proven unreachable in the case of manual cleaning. This feat was achieved by removing the suspended gondolas, wires and ropes that were mandatory in traditional façade glass cleaning, instead using different types of locomotion and adhesion mechanisms that facilitate the movement of the robot on the glass.

The current research interest in developing various automated wall-climbing robots capable on maneuvering on vertical or inclined surfaces have increased the productivity and efficiency in cleaning operations, that have determined a considerable rise in the production of said robots.

While this advancement is putting in danger the considerable amount of classic façade glass cleaning businesses that have flourished over the years, it is also considered an immense improvement in the work safety area, while also pushing existing businesses to adopt the automatization of the working environment.

Even if the cost of said advancements are more considerable than the cost of the working force in the case of manual cleaning, we need to take into consideration that a automatized solution is more profitable on the long term, because it eliminates the additional costs deducted from the training of personnel, while also being more time efficient and risk free.

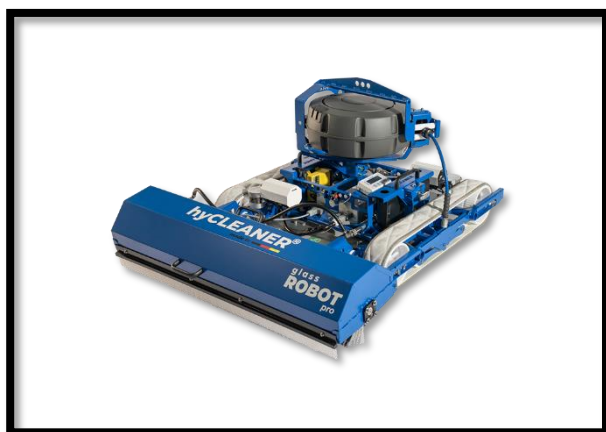


Figure 1.5 Example of robotized solution

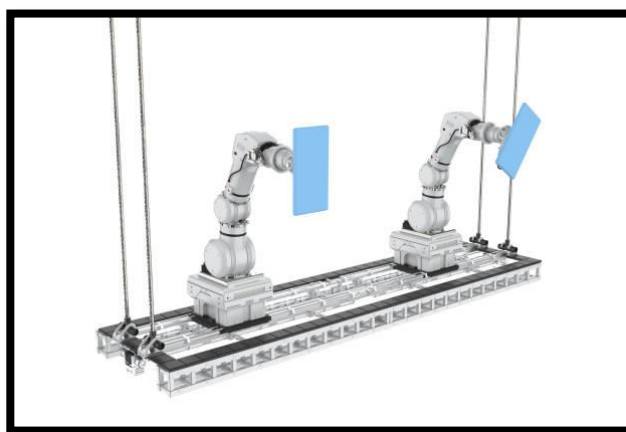


Figure 1.6 Another example of a robotized solution

1.2.1. Semi – automated VS fully automated robotized solutions

As we discussed before, when talking about robotized solutions we have two candidates. Each of them is proposing similar solutions in cleaning the glass of façades, the main difference being in technologies used, mechanical structure and most importantly on the degree of automatization.

Semi-automated solutions – are usually controlled by a worker from a close distance, using a navigation device in order to control the movements of the robot. The worker in cause is usually less than 20 meters away, in a mobile scaffold at the same attitude as the robot, on the roof of the building, or in the said building, while the movements are transmitted to the robot wirelessly. The limited range of communication between the robot and the worker explains why he has to be close to the robot.

The robot is tethered by a security chord to the building or to the scaffold in which the worker controls it for security measures, in case the robot's adhesion mechanism fails and it detaches from the façade.

Fully automated solutions – on the other hand, do not require any additional assistance from a human operator, while working. In the same manner as the semi – automated variants, it is tethered to the building using ropes, wires or scaffolds. They are equipped with sensors and cameras that collect information of the surrounding in order to create 3D maps of the working environment, while also calculating it's every movement for the sake of a high efficiency in cleaning.

The behavior of the robot can be tracked through a GUI – graphical user interface, by an operator who can make decisions depending on the progress of the robot or if a problem arises.

If we are talking about either semi-automated or fully automated robotized solution, each and one of them have some advantages and disadvantages, coming down at the cost of manufacturing, cost of implementing the solution while also considering the possibility of said implementation in a real-life scenario, and also the cost of maintenance, in addition to the facts underlined in the lines above.



Figure 1.7 Semi-automated robot. hyCleaner glassROBOT pro



Figure 1.8 Fully automated robot. Ozmo developed by Skyline Robotics. KUKA

RESULTS FROM 1ST SEMESTER

2. Process of Thought

The first step in this project would be to underline the main components that are taken into consideration when designing and implementing the robotized solution for cleaning glass façade. While taking into consideration the notions discussed in the first chapter, we have the following components:

- a) The mechanical structure of the robot
- b) The technologies used in automating the cleaning process
- c) The total costs of designing and implementing of the solution

With these three factors in mind, we can understand them better by analyzing currently developed solutions over the world, as a point of start in our study.

2.1. Semi – automated robots

One such solution for semi-automated robots is produced by hyCLEANER, in Germany. As explained in the previous chapter, this is the kind of robot that is controlled remotely by an operator that has to be near the robot. A cord thethers the robot to the building or a near structure, while the worker is controlling the path taken.

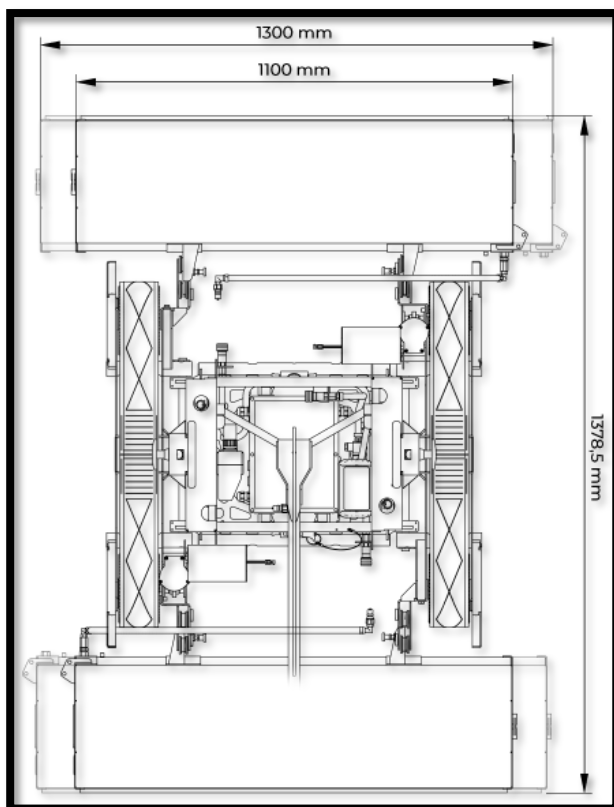


Figure 2.1 Dimensioning for the hyCLEANER robot

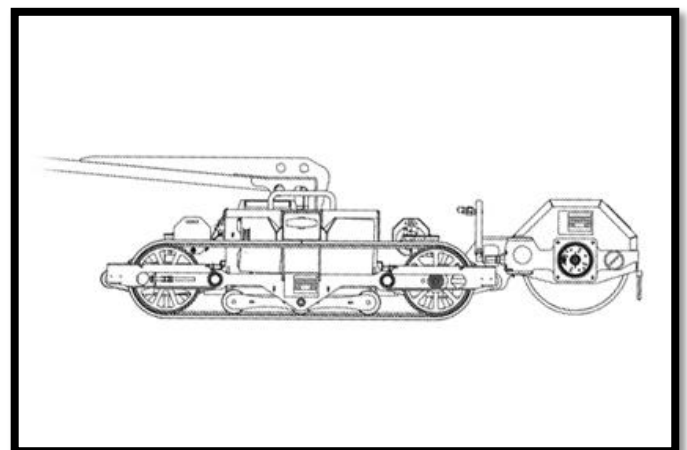


Figure 2.2 Side view of the robot

As mentioned by the producer, the cleaner is ECO- friendly because of the cleaning solution that involves a reduced amount of water, approximately 8 liters per minute, which consists 20 times less than conventional high – pressure cleaning systems.

The system is easy to control and easy to learn making the training phase less time consuming than with other products. The flexibility of the product comes from the fact that it can be used on different types of glass while also being easy to disassemble and transport.

The producer mentions the fact that the robot can clean up to 2400 square meters per hour.

After further inspections, we deduce that the cleaning solution is composed of a cylindrical roll with fine bristles that rotates, while being constantly lubricated with water.

The moving mechanism is composed from two rows of tracks, similar with the solid chain tracks used for tanks, denoting the fact that this robot is not able to move on surfaces that are steeper than 30 degrees, because of the risk of falling.

Further modification could be added to the robot, by adding an adhesion mechanism that could allow it to move on the facade of buildings. Outside of this the robot is equipped with sensors used for edge detection, so the robot does not fall off the roofs.

2.2. **Fully automated robots**

In the case of fully automated robots we have a variant that is very similar to the semi automated solutions presented before, the difference being in the fact that it does not need to be operated by a human. One of such robots is the kite robot developed by Kite Robotics, which consists of a central cleaning element, a robot equipped with an identical cleaning solution as the one that hyCLEANER proposes.

The robot is attached to the four corners of the facade by four light-weight wires which navigate by specially developed winches, on the corners.

By changing the wire lengths in a coordinated way, the robot is able to move itself and clean the facade in vertical adjacent tracks.

Once the robot is connected, it calibrates itself and automatically starts cleaning in such a manner that after it finished cleaning, the robot returns to its starting position, where the system can be easily disconnected again. The robot is proficient in cleaning surfaces that have up to 20 cm depth, but nothing more than that.

The flexibility of the robot comes from its automated cleaning program that is a pre-programmed route, while also being able to be controlled remotely, becoming a semi- automated cleaning robot.

The maintenance of the robot is optimized and calculated from a database that is collecting in real time information regarding the robot running time and motor output that is collected from the sensors equipped on it.

The producer mentions that the cleaning robot solution is capable of covering 300 square meters of glass per hour.



Figure 2.3 The kite robot working concept



Figure 2.4 The tethering solution of the robot

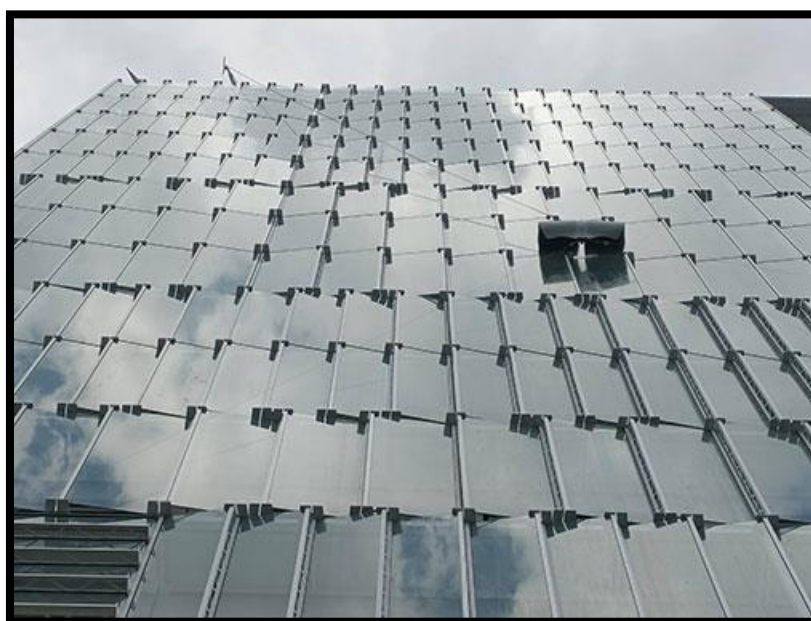


Figure 2.5 The kite robot in action

Another solution proposed in the case of automated robots is a very innovative one simply because it is the only functional robotization, currently on the market.

It is about the project Ozmo: Kr Agilus, developed by Skyline Robotics, a special project developed by KUKA that is focused on developing and upgrading existing solution in the business of facade cleaning.

The uniqueness of the solution proposed by Skyline Robotics comes from the fact that they use robotic arms that clean the facade glass through the use of cleaning brushes attached as end – effectors.

The solution is composed of two of these robots that are fixed on a scaffold which moves with the robot. Basically it is a system that combines KUKA's Kr Argilus robots with a computerized vision system, artificial intelligence and machine learning technologies.

In order to provide the power and water needed in order to execute the cleaning tasks, Ozmo is designed to fit into a building's existing building maintenance unit infrastructure, gathering the power and water supply from the building itself.

Within the scaffold basket, a table houses several types of sensors and computers used to communicate with the robot and its other utilities.

The robot arm is equipped with a Lidar camera that uses class 1 lasers for imaging, a category of lasers safe for humans. The system is ready to work once the table its fixed on the mobile scaffold.

The robot arm is continuously communicating with the Lidar camera that collects surrounding images and creates a 3D map of the working area. In combination with other sensors, these technologies are used to continually optimize the path planning at a rate of about 200 times per second, while automating the robot's descent down the glass of the facade.

As mentioned before, the robotic arm used in this solution is the KUKA Kr Agilus, a 6 axis robot that is built to sustain harsh environments, while being totally waterproof.

Skyline has developed an algorithm that allows for feedback for force control and other variables that give the possibility to control how much pressure the robot applies to the glass, through the use of force torque sensors, placed on the cleaning brush device.

Through the use of artificial intelligence, the robot can stabilize in real time and recalculate the cleaning path, while taking into consideration different environmental hazards that can modify it, like the wind blowing or peculiarities of the mobile scaffold kinematics while ascending and descending. The continous path adjustment is detrimental in figuring out different cleaning paths that need to be approached when the scaffold of the building is at a certain angle or presents different obstacles.

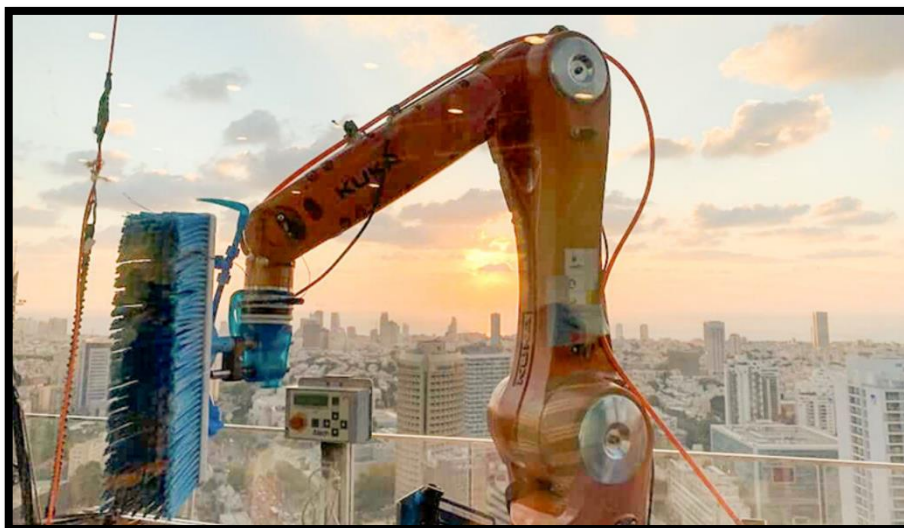


Figure 2.6 Cleaning process as seen from the inside of the building



Figure 2.7 The Ozmo system in action. Automated facade glass cleaning



Figure 2.8 KUKA KR Agilus 6-axis robotic arm

3. Considering the current solutions

After taking into consideration the current development of the facade cleaning robots and other sources, we need to address the solution for each component that is part of the final robotized assembly. With this being said, we have the following:

3.1. Locomotion mechanisms

The optimal movement of the robots is secured by the locomotion mechanism of the said robot, or of the container/compartment where the robot is secured. As such, the most common locomotion techniques are mostly similar to the ground mobile robots. The track like structure composed of multiple wheels that are engaged by a caterpillar track can represent a good solution, especially if the track structure is fabricated out of a soft material that facilitates the movement on glass, without scratching it. This way silicone enveloped wheels and a silicone caterpillar track can represent a good solution.

Translation and sliding frames or guide rails can also represent a solution for more precise movement, while also eliminating the need for an adhesion mechanism that needs to keep the robot in contact with the glass while moving.

The problem with this kind of solution is the additional preparation that needs to be made to the building that needs to be cleaned. Such installment needs proper time and is very time consuming and costly when talking about the need to maintain it.

The best solution would be a much complex one that would involve the concept of nested reconfigurable robots. The concept aims to develop robots that can be applied to autonomous facade cleaning on buildings with various types of architecture, such as rounded glass surface, spherical surface, or different types of structures that can limit the movement of the robot or restrict it at all, like beams or reams. Because the typical facade cleaning robots work on flat glass panels that are connected by frames, they need to clean the windows according to the frame shape.

The concept achieves autonomous cleaning according to window shapes, while using multiple modular multilegged robots capable of reconfiguring their morphology based on window shapes, which is executed by transforming their own module or connecting with each other and letting the robots cooperate.



Figure 3.1 Caterpillar rail solution

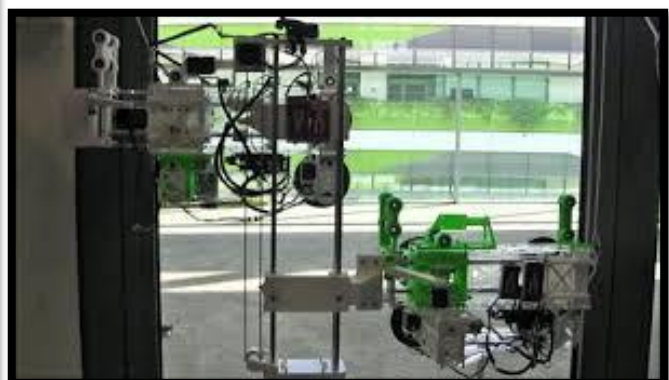


Figure 3.2 Nested Reconfigurable robots concept

3.2. Adhesion techniques

The adhesion techniques are mandatory for the robot, as without it, it is not possible to keep itself attached to the glass without additional support structures that can hinder or restrict its movement on the facade in the scope of cleaning the glass.

As such, there are a few techniques adopted in the development such as pneumatic or magnetic solutions. The magnetic solution is realised with the use of permanent magnets, electromagnets or even electro – permanent – magnets that are attached to the robot. This solution suffers from the same disadvantage as the locomotion mechanism that involves rails.

There are some less common solutions that involve electro or chemical adhesion.

Out of all of them, the pneumatic adhesion is the most used and suitable for developing facade cleaning robots, particularly with the high payload. The combination of active and passive pneumatic adhesion generated by the active vacuum of the suction cups is the real selling point of this solution. The active vacuum is generated by a solenoid valve, an electromechanical valve that can control the airflow in a vacuum system to help create or dissipate the vacuum.

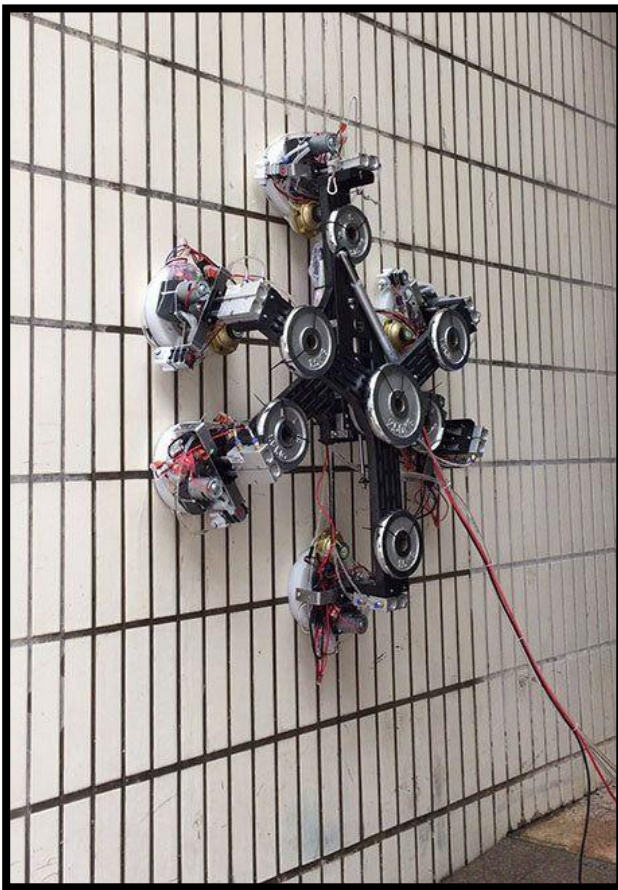


Figure 3.2 Suction cups in action for a climbing robot



Figure 3.3 Solenoid valve

3.3. Instruments for data collection

When we talk about data collection we are talking about the way of the robot to communicate with the surrounding in order to determine the cleaning pattern it needs to follow while also making micro-adjustments constantly, and maintaining the efficiency and safety during the entire work process.

First of all we need a device for image acquisition, and for this task, the Lidar camera is the best candidate, as explained in the previous chapters. Through it we can collect very precise data regarding to the surface that needs to be cleaned and the nearby surroundings so the robot can readjust its trajectory in real time, especially if it is close to one of the building edges.

Pressure force sensors are also important for controlling the force that is applied by the robot's cleaning mechanism, so that the glass is not scratched or cracked. This kind of sensors in combination with vacuum regulators can be used in order to maintain a consistent level of vacuum by adjusting the flow of air. This is a crucial factor because in order for the robot to move on the glass, the vacuum inside the adhesion mechanism needs to be weak enough for the robot to slide freely on the facade, but strong enough in order for it to remain attached to it.

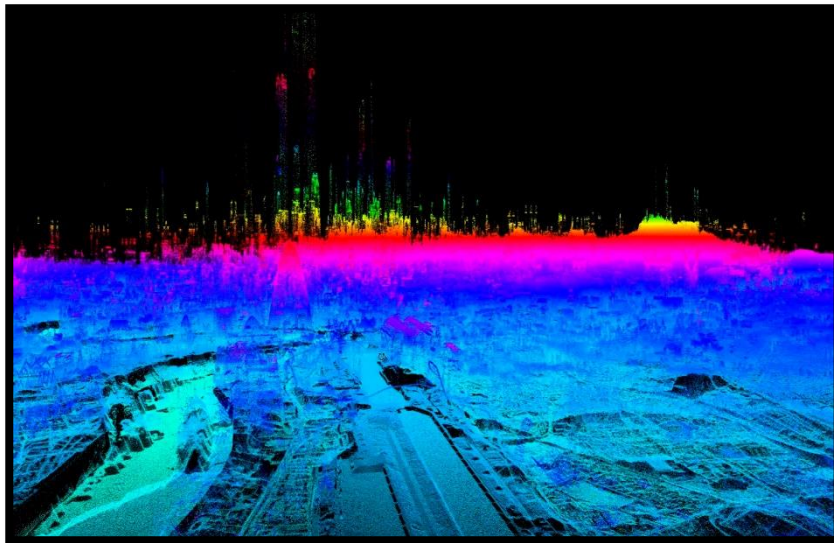


Figure 3.4 Lidar data visualisation. Depth and distance representation



Figure 3.5 A lidar camera developed by intel

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2ND SEMESTER

ADJUSTMENTS TO THE WORK PLAN

1. OBJECTIVES AND ACTIVITIES ADJUSTED FOR SEMESTER 2 IN RELATION TO THE INITIAL PLAN

The second semester has as main objective to revisit the topics discussed in the previous semester, review them and make adjustments.

A second objective would be to prioritise the most important functionalities of the robot and start implementing them. A important thing would also be to start a rough 3D model of the robot.

2. ADJUSTED OBJECTIVES AND ACTIVITIES FOR SEMESTER 3 IN RELATION TO THE INITIAL PLAN

The third semester would resume the implementation of the rest of the functionalities while also re-shaping the 3D model of the robot.

3. ADJUSTED OBJECTIVES AND ACTIVITIES FOR SEMESTER 4 IN RELATION TO THE INITIAL PLAN

The plan for the last semester remains unchanged.

RESULTS FROM 2ND SEMESTER

4. Implementation of the adhesion solution

Turning back to the adhesion solutions discussed in the previous semester, we have decided that a pneumatic system would be the best choice for the robotised facade cleaning mechanism.

Upon further investigation, in order to implement such a solution we need to take into consideration additional key components and techniques to ensure the reliability of the adhesion mechanism.

4.1. Overview of the methods and components required for implementation

4.1.1. Vacuum Pump Control

A vacuum pump is essential for creating the vacuum necessary for adhesion. The control of the vacuum pump can be achieved through:

- Variable Speed Pumps: Using a variable speed pump allows for precise control of the vacuum pressure. By adjusting the speed of the pump, the pressure can be increased or decreased as needed.
- On/Off Control: In simpler systems, the pump can be controlled by turning it on and off based on pressure readings. This can be done using a relay or a transistor switch controlled by a microcontroller.

4.1.2. Pressure Sensors

Pressure sensors are used to monitor the vacuum level. These sensors can provide real-time feedback to the control system, which can then adjust the pump operation accordingly.

- Analog Pressure Sensors: Provide continuous readings of the vacuum pressure.
- Digital Pressure Sensors: Offer discrete pressure values and can be easier to interface with digital control systems.

4.1.3. Control Algorithms

The vacuum pressure can be controlled using various algorithms implemented in a microcontroller or a PLC (Programmable Logic Controller):

- PID Control: Proportional-Integral-Derivative (PID) control is a common method for maintaining stable vacuum pressure. The PID controller adjusts the pump speed or the valve position based on the error between the desired and actual pressure.
- On/Off Control with Hysteresis: This simpler control method involves turning the pump on when the pressure falls below a certain threshold and off when it rises above another threshold.

4.1.4. Valves

Valves can be used to control the flow of air in and out of the vacuum chamber. As discussed in the previous semester we could use these valve types:

- Solenoid Valves: Electrically controlled valves that can open or close to adjust the vacuum pressure.
- Proportional Valves: These allow for more precise control by adjusting the valve position in response to control signals.

4.1.5. Sealing and Suction cups

Ensuring a good seal between the robotic system and the wall is crucial for maintaining vacuum pressure. This involves:

- Suction Cups: Made from materials like rubber or silicone, these need to be properly sized and positioned to ensure a good seal.
- Sealing Mechanisms: Gaskets or other sealing materials around the edges of the suction cups can help maintain the vacuum.

4.1.6. Redundancy and Safety

For safety and reliability, redundancy can be built into the system:

- Backup Pumps: Having an additional pump can ensure that the vacuum is maintained even if one pump fails.
- Check Valves: These prevent the backflow of air into the vacuum chamber in case of a pump failure.

4.2. Example of a control loop

1. **Initialization**: The system powers on, and initial pressure readings are taken.
2. **Pump Activation**: The vacuum pump is activated, and pressure readings are continuously monitored through the use of digital pressure sensors. I had chosen the digital version of the sensors because they are easier to implement than the analog counterpart, eliminating the need of an Analog-to-Digital converter while also being less susceptible to noise compared to analog readings which results in more stable readings.
3. **Pressure Monitoring**: The pressure sensor provides real-time feedback to the microcontroller.
4. **PID Control**: The microcontroller adjusts the pump speed based on the PID algorithm to maintain the desired vacuum pressure.
5. **Valve Adjustment**: If needed, solenoid or proportional valves are adjusted to fine-tune the pressure.
6. **Safety Checks**: Regular checks are performed to ensure the vacuum is within safe operating limits, and backup systems are ready to engage if needed.

For a better understanding of the vacuum mechanism we have representative scheme as seen in the figure 4.1.

The pressure of the compressed air generated by the compressor is reduced by the regulator, after which it is filtered. The air flows through the solenoid valve and inside the vacuum generator, where the air from the atmosphere is also sucked in through the vacuum pad. Before reaching the generator, it passes through a filter after which it is released with the compressed air back into the atmosphere.

Filters are mandatory for ensuring a good functionality of the system loop while also prolonging the lifetime of it by filtering the impurities in the air.

The use of silencers is optional in this application because noise suppression does not represent a main target.

The vacuum switch is an electronic or mechanical device that provides an electrical switching contact function at a defined vacuum gas pressure level. It's used for pressure control purposes and safety interlock functionality.

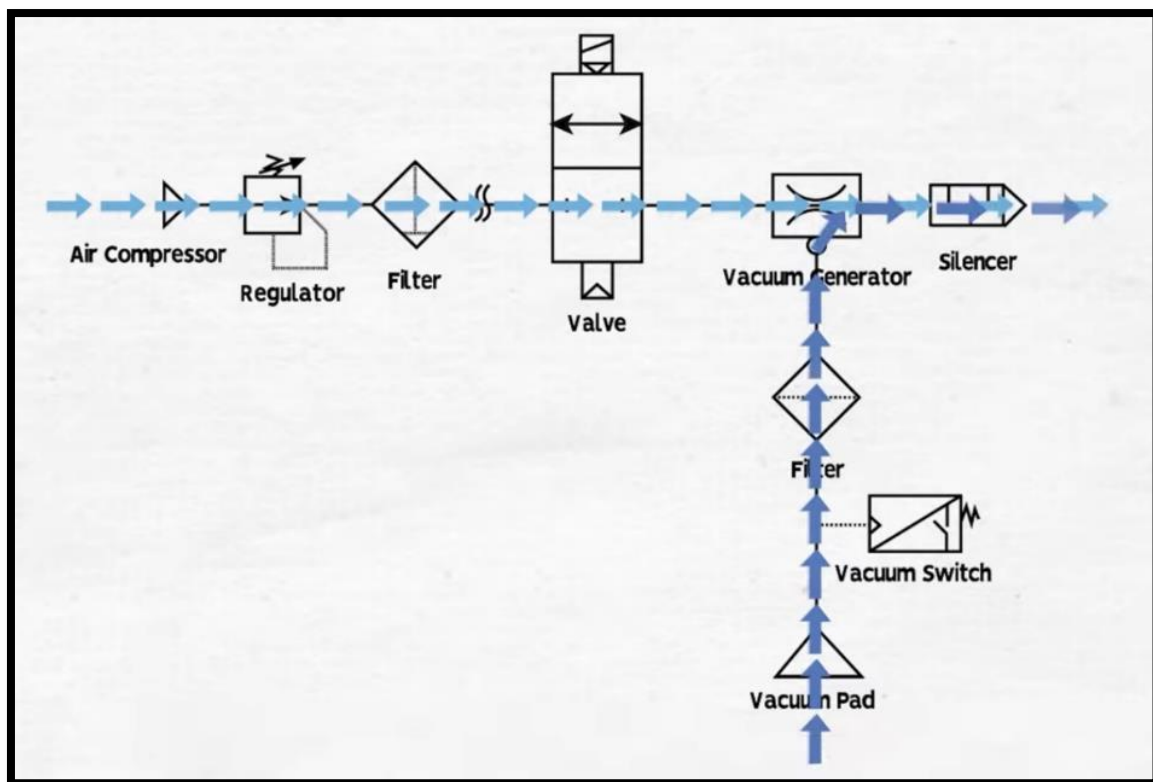


Figure 4.1. Vacuum loop schematic. Air flow inside the mechanism: Light blue- compressed air, Dark blue – atmosphere air.

4.3. Hypothetical PID implementation

Considering the previous statements regarding the control loop of the adhesion mechanism solution, we also have to come up with the implementation of the PID control. Taking this into consideration we have the following string of code:

```
import time
import PID
import digital_pressure_sensor

# Setup PID controller
pid = PID.PID(1.0, 0.1, 0.05)
pid.SetPoint = desired_vacuum_pressure
pid.setSampleTime(0.1)

# Initialize digital pressure sensor
sensor = digital_pressure_sensor.Sensor()
sensor.initialize()

while True:
    current_pressure = sensor.read_pressure()
    pid.update(current_pressure)
    control_signal = pid.output

    # Control vacuum pump speed
    set_pump_speed(control_signal)

    # Safety check
    if current_pressure < min_safe_pressure or current_pressure > max_safe_pressure:
        engage_backup_system()

    time.sleep(0.1)
```

Firstly we are importing the PID control library, which provides the necessary functions and classes to implement a PID controller. The PID library is assumed to be a hypothetical or custom library.

Then we create the PID controller instance where **PID.PID** is the constructor for creating an instance of the PID controller. The numbers **1.0**, **0.1**, and **0.05** are the PID controller parameters:

- **1.0** is the proportional gain (K_p).
- **0.1** is the integral gain (K_i).
- **0.05** is the derivative gain (K_d).

These parameters determine how the PID controller will respond to the error (the difference between the setpoint and the measured value).

In the following line we are setting the desired setpoint where **pid.SetPoint** sets the target value that we want to maintain. In this case, it is **desired_vacuum_pressure**, which represents the required vacuum pressure for the robot.

desired_vacuum_pressure should be defined accordingly and represents the pressure level that the PID controller will try to achieve and maintain.

After that we need to set the sample time of the pressure. This way **pid.setSampleTime(0.1)** sets the interval (in seconds) at which the PID controller will calculate the control output. Here, it is set to 0.1 seconds (100 milliseconds) as a starting point. This means the PID controller will update its calculations and adjust the control signal every 0.1 seconds.

4.3.1. Summary

- **PID Initialization:** Setting up the PID controller with specific gain values.
- **Setpoint:** Defining the target vacuum pressure to be maintained.
- **Sample Time:** Defining how often the PID controller should update its calculations.
- **Control Loop:** Continuously reading the current pressure, updating the PID controller, adjusting the pump speed, and performing safety checks.

4.3.2. Explanation of PID Control

Proportional (P): This part of the controller reacts proportionally to the current error. If the error is large, the proportional response will also be large. The proportional gain (K_p) determines the strength of this response.

Integral (I): This part of the controller sums the past errors to eliminate residual steady-state errors. The integral gain (K_i) determines how strongly the integral term reacts to accumulated errors.

Derivative (D): This part of the controller predicts future error based on its rate of change. The derivative gain (K_d) determines the strength of this prediction, helping to dampen the system's response and reduce overshoot.

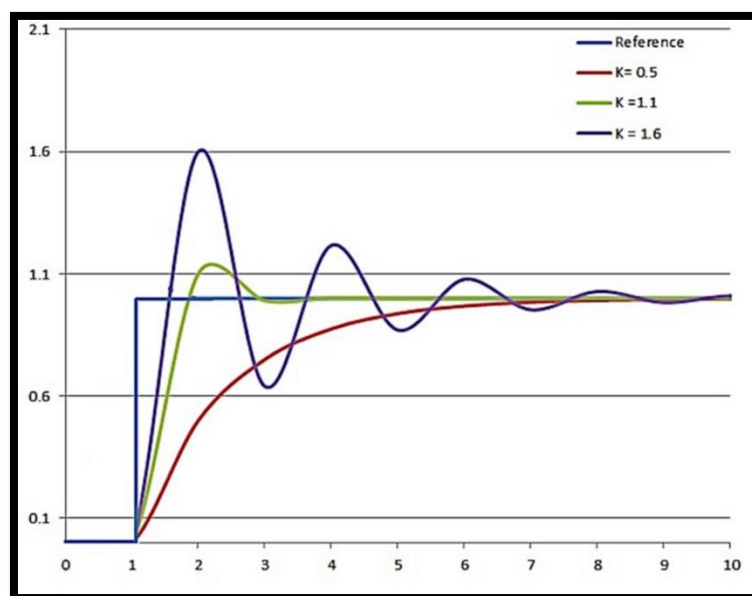


Figure 4.2. Graphical representation of the PID variables where the green line represents the P-proportional, red is I-integral, blue is D- derivative

5. Implementation order of the systems and the importance of it

5.1. Analytic Hierarchy Process chart

For a better prioritization of the systems that need to be implemented in order for the robot to work we will create a analytic hierarchy process chart.

On the left side, on yellow background, we have the ten most important attributes of the robot. Below that we have the prioritization matrix where we compare the importance of the robot's attributes with eachother.

If the left attribute is more important than the top one, then a score is written down depending on how high or low is the difference. The legend is located above the matrix.

Based on the prioritization matrix, a chart is created where the order of importance is highlighted, while also presenting the three most important attributes.

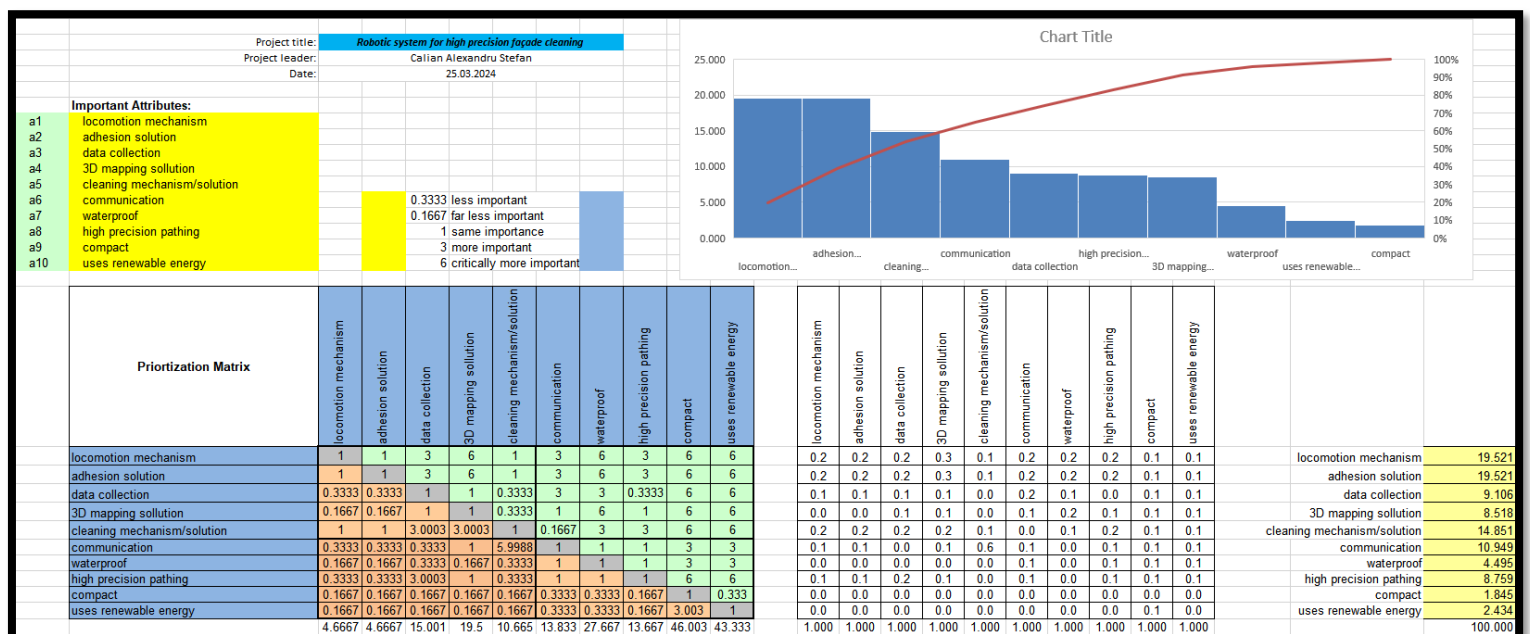


Figure 5.1. Analytic hierarchy process chart. Prioritization matrix and the first ten most important technological attributes of the robotic system.

5.2. Quality Function Deployment chart

This chart was created in order to better understand the need of the client and not on what the company may believe the customer desires.

On the left side, on yellow background, we have the main functions of the robot co-related with the ten main attributes set in the AHP. Below is the relationships matrix where the customer requirements are compared with the functional requirements.

Depending on the results a chart is generated on the top side of the document where the importance of the functionalities is highlighted from the perspective of the client, while also presenting the four most important ones.

The order in which each feature of the high precision facade glass cleaning robot is tackled is going to be defined by these charts.

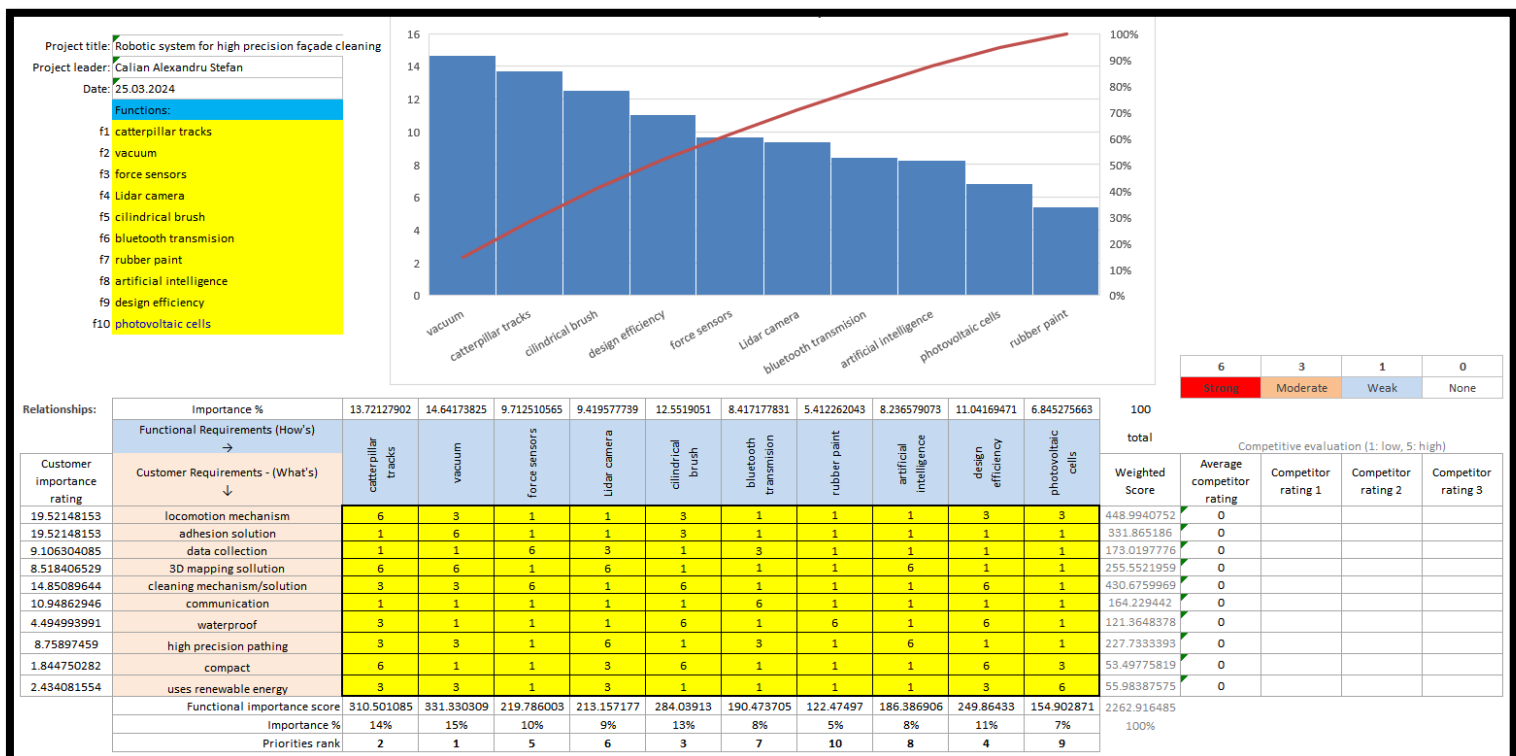


Figure 5.2. Quality Function Deployment chart. Relationships between customer requirements and functional requirements.

6. Implementation of the cleaning solution

Based on the previous semester research we have come to the conclusion that a microfiber roller solution will constitute the cleaning mechanism of the robot. Following this we need to follow the main steps in implementing it.

6.1. Design requirements and specifications

First of all we need to determine the volume of water required based on the area that needs to be cleaned and the robot's operational duration.

We also have to decide the appropriate flow rate for water dispensing in order to ensure an efficient cleaning.

6.2. Components Selection

An reliable pump needs to be selected so that it can handle the required flow rate and pressure of the water.

For this particular situation we can choose a peristaltic pump paired with electronically controlled valves (used in regulating the flow of water) because of its high precision liquid dispensing without contaminating the water and also because of its low maintenance, factors that are vital for this application.

Another important component would be the nozzle, through which the water is dispensed on the roller. The nozzle should be able to produce a fine mist or a targeted wide spray that would coat the roller uniformly.

6.3. Control System

The pump and valves should be connected to the microcontroller of the robot. After that the software necessary to regulate the dispensing of water and the flow rate of it allowing real-time adjustments through the use of sensors that give feedback consistently.

6.4. Software Development

Control algorithms will need to be developed in order to control the timing and amount of water dispensed based on the position of the robot and cleaning requirements.

Safety features are also an important feature that needs to be implemented for leak or blockage detection or pump failures, while also halting the water dispensing process if necessary.

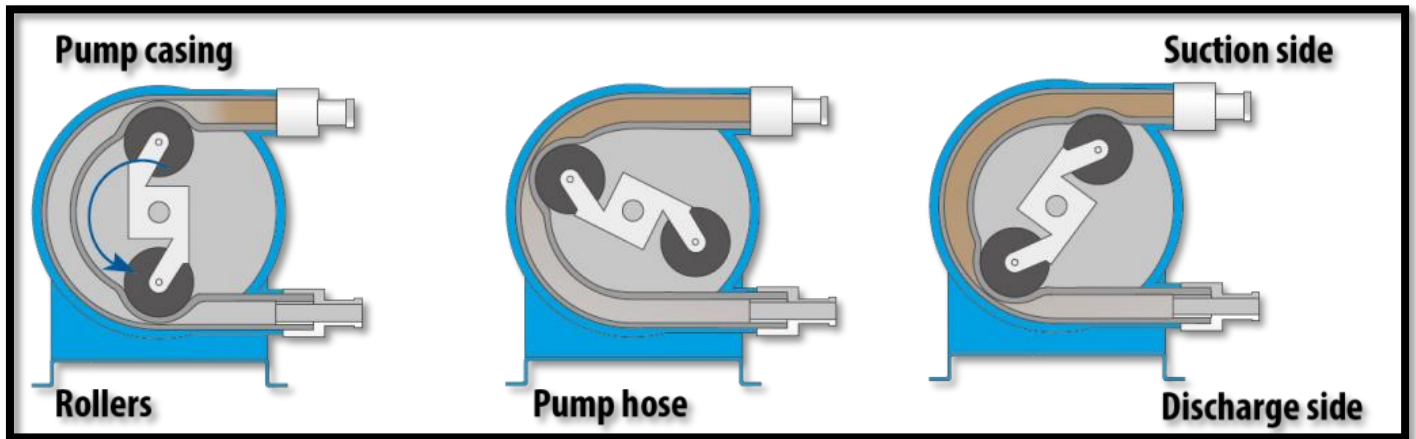


Figure 6.1. Visual representation of a peristaltic pump. The working concept of a peristaltic pump.



Figure 6.2. Similar working concept of the water nozzle mentioned in the application.

7. Starting prototype of the robotic system

In order to offer a better view of the robotic solution, i created a rough 3D model of the cleaning robot. This model is only made in order to represent the main functionalities of it while also creating a starting point for the design phase of the next semester.

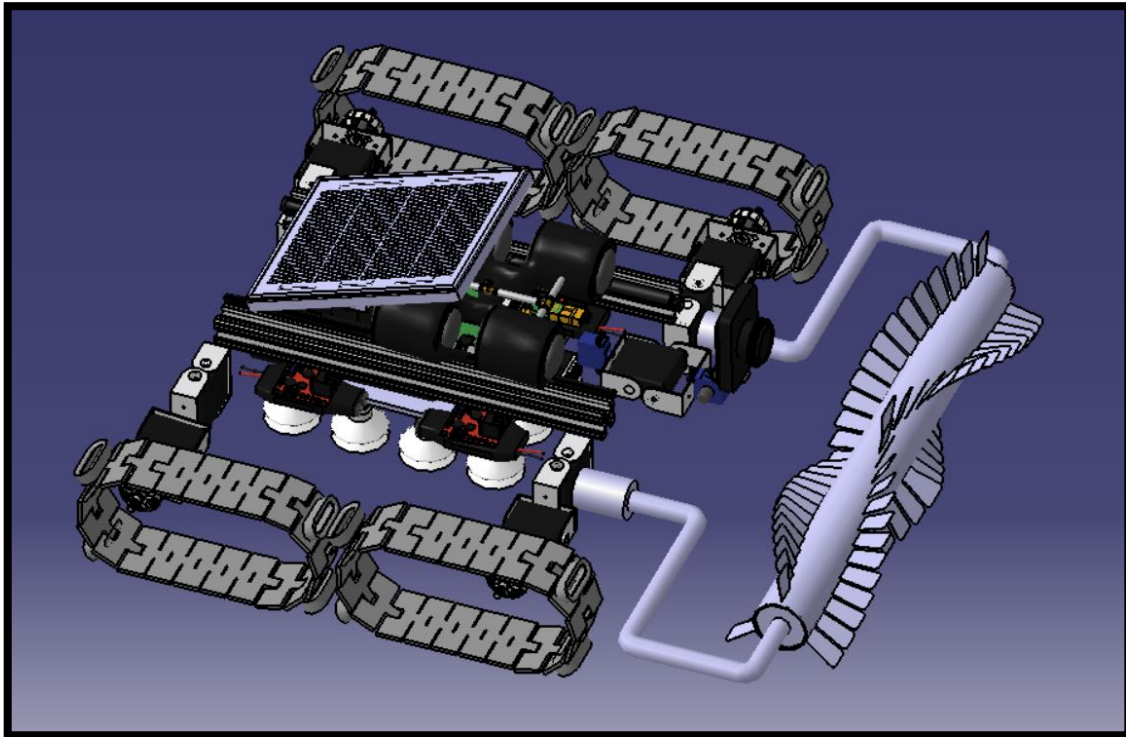


Figure 7.1. Isometric view of the robotic system for high-precision glass - facade cleaning

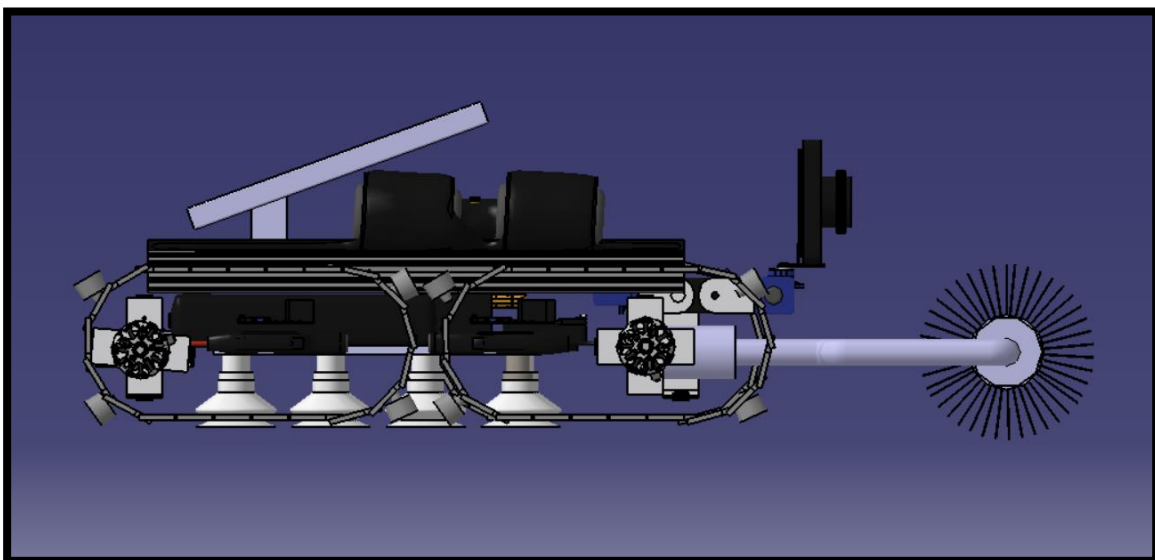


Figure 7.1. Side view of the robotic system for high-precision glass - facade cleaning

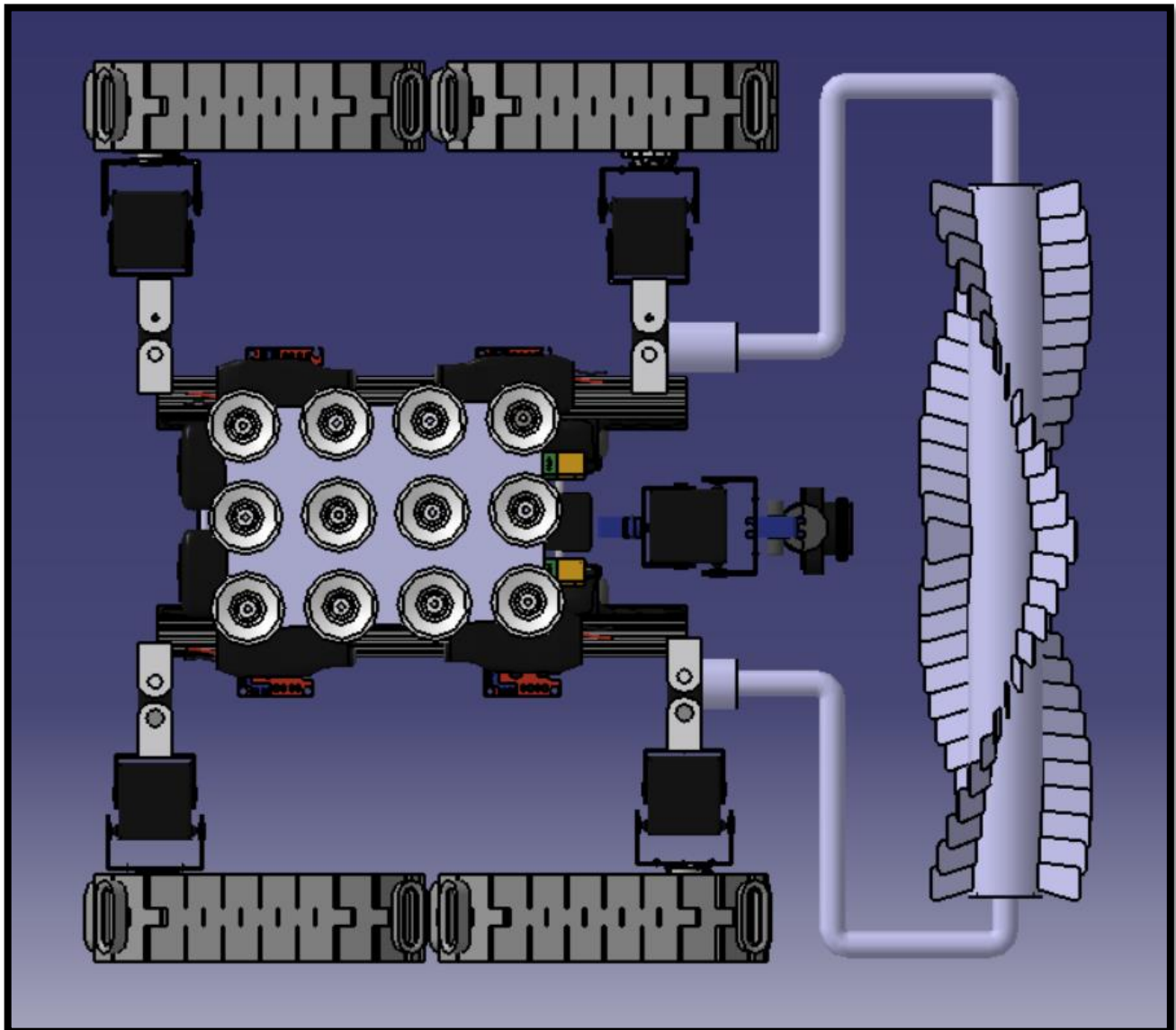


Figure 7.1. View from below of the robotic system for high-precision glass - facade cleaning

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3. „How Vacuum Generators work?” - <https://www.youtube.com/watch?v=l-wZYJXpVFg>
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5. „Introduction to Quality Function Deployment (QFD)” - <https://quality-one.com/qfd/>

3RD SEMESTER

ADJUSTMENTS TO THE WORK PLAN

1. ADJUSTED OBJECTIVES AND ACTIVITIES FOR SEMESTER 3 IN RELATION TO THE ORIGINAL PLAN

Integrating the rest of the robot system functionalities.

2. ADJUSTED OBJECTIVES AND ACTIVITIES FOR SEMESTER 4 IN RELATION TO THE INITIAL PLAN

The plan for the last semester remains unchanged.

RESULTS FROM 3RD SEMESTER

8. Implementation of the data collection solution

Resuming the implementation of the other solutions needed in configuring the robotic system, we have to implement a data collection system, which involves capturing relevant data to improve performance, ensure safety, while also providing insights for maintenance and optimization.

8.1. Defining the objectives in the data collection

First of all we need to identify what kind of data is needed and why. For that matter we have:

- Operational Data: Needed in order to track robot movements, cleaning patterns, and time per unit area.
- Environment Data: With the use of it we measure weather conditions like wind speed and humidity while also taking into consideration the surface conditions like dirt level and the surface tension.
- Performance Metrics: Evaluate cleaning effectiveness, precision, and efficiency.
- Safety Monitoring: Used in monitor tilt angles, proximity to edges, and collision detection.

8.2. Selection of sensors and hardware

Next we have to consider the appropriate sensors for data collection with which the robot should be equipped:

- Positioning and Navigation: Done by using GPS, LiDAR, or IMUs for tracking the robot's position and trajectory.
- Cleaning Feedback: Done through the use of cameras, infrared sensors, or optical systems to analyze cleaning quality.
- Environment Monitoring: Includes sensors for wind speed, temperature, and humidity.
- Load Monitoring: With the use of strain gauges and accelerometers we can detect excessive loads or anomalies.
- Edge Detection: By using proximity sensors or ultrasonic sensors we are able to ensure safe operation near edges.

8.3. Designing a data processing architecture

In this case we need to implement a system for processing and storing collected data:

- Onboard Processing: Using microcontrollers or edge computing devices like Raspberry Pi or NVIDIA Jetson to preprocess data.
- Real-time Monitoring: Transmitting critical data like safety metrics in real-time to a remote monitoring station.
- Cloud Integration: Storing historical data in a cloud database for analysis and long-term insights.
- Data Reduction: Compressing or filtering data to reduce bandwidth and storage requirements.

8.4. Implementing the communication frameworks

This is done in order to ensure a proper communication for data transfer:

- Wireless Connectivity: Using Wi-Fi or 4G/5G for real-time data transmission.
- Data Formats: Using standardized formats like JSON, MQTT for interoperability.
- Redundancy: Design fallback systems like a local storage to handle the communication failures.

8.5. Development of software systems

Further along we have to create software for data collection, processing and visualization:

- Data Collection Software: The implementation of firmware to integrate sensor data and manage device operations.
- Data Analytics: Using machine learning models or statistical tools to analyze performance trends and predict maintenance needs.
- User Interface: That provides a dashboard to visualize cleaning patterns, operational efficiency, and maintenance alerts.

8.6. Power and durability

We have to ensure that the system is reliable and power-efficient:

- Battery Management: We integrate efficient power supplies and battery monitoring systems.
- Weatherproof Design: Using waterproof and vibration-resistant components.

8.7. Incorporating safety and redundancy in the system

We have to implement fail-safes and robust systems:

- Redundant Sensors: Using multiple sensors for critical measurements to cross-verify data.
- Error Detection: Building error-handling algorithms to identify anomalies in real time.
- Emergency Systems: Including mechanisms to halt operations or trigger alarms when safety thresholds are breached.

In addition to the solutions mentioned before we have to conduct thorough testing while also ensuring long-term functionality and adaptability for the robotic system by adopting the following ideas:

- Simulations: By using virtual environments to simulate various conditions and validate the system.
- Field Tests: Testing the robot in real-world scenarios, collecting data for validation and calibration.
- Feedback Loop: Using insights from data analysis to refine the robot's algorithms and design.
- Predictive Maintenance: By using data trends to predict failures and schedule maintenance proactively.
- Modularity: Designing the system to easily integrate new sensors or software updates as technology evolves.

Below we have a example workflow for all the steps mentioned before:

1. Sensors collect data like wind speed, cleaning pressure.
2. Edge Device preprocesses and sends key metrics.
3. Cloud Database stores detailed logs for analysis.
4. Analytics Software visualizes cleaning coverage and predicts equipment wear.
5. Dashboard provides real-time operational insights to operators.

A proper visualization of the workflow is presented in the figure 8.1:

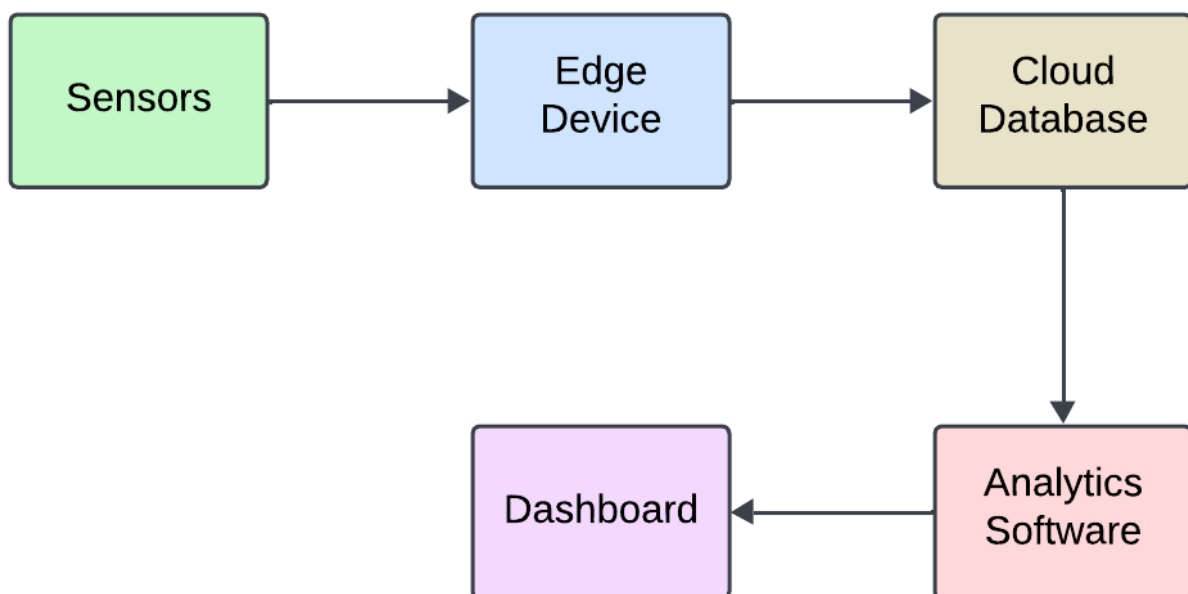


Figure 8.1. Workflow diagram of the data collection system

9. Calculating the gripping force of the suction cups

In order to achieve a good adhesion, the suction cup needs to have a high gripping force while at the same time, it should not be too high so that we can reduce the torque force of the motors that move the caterpillar tracks.

For the forward movement, the robot will use linear actuators which need to have the capacity to push or pull the robot system weight. As so, the calculation is done to ensure the right choice of suction cups and motors are chosen for the robot gripping and locomotion mechanism.

9.1. Horizontal lifting force

Considering that our robot is on the smaller size and it's also made using lightweight materials like aluminum and carbon fiber, we have a general weight of about 10 kg.

By applying Newton's Law to calculate the force on the 10 kg robot with a change in acceleration of 2 m/s^2 and a safety factor of $S_H = 2$

$$\begin{aligned} F_H(N) &= \text{mass}(kg) * (a_g + a) * S_H \\ F_H(N) &= 10kg * (9.81\text{m/s}^2 + 2\text{m/s}^2) * 2 \\ F_H &= 236.2 \text{ N} \end{aligned}$$

9.2. Vertical lifting force

By applying the Newton Law to calculate the force on a 10 kg robot with a change in acceleration of 2m/s^2 and a safety factor of $S_V = 4$

$$\begin{aligned} F_V(N) &= \text{mass}(kg) * (a_g + a) * S_V \\ F_V(N) &= 10kg * (9.81\text{m/s}^2 + 2\text{m/s}^2) * 4 \\ F_H &= 472.4 \text{ N} \end{aligned}$$

Calculating the force on a 10 kg mass with a dry surface, and a change in acceleration of 2m/s^2

$$\begin{aligned} F_M(N) &= \sqrt{(F_V^2 + F_H^2)} \\ F_M(N) &= \sqrt{[(10kg * 2\text{m/s}^2) * 4]^2 + [10kg * (9.81\text{m/s}^2 + 2\text{m/s}^2) * 2]^2} \\ F_M(N) &= \sqrt{(80kg\text{m/s}^2)^2 + (236.2kg\text{m/s}^2)^2} \\ F_M(N) &= \sqrt{6400kg\text{m/s}^2 + 55790.44kg\text{m/s}^2} \\ F_M &= 249.38 \text{ N} \end{aligned}$$

At 90% of full vacuum and 12 suction cups working simultaneously, the area of the suction cup used can be calculated. Since the robot is using 12 suction cups in order to hold onto the window pane, as mentioned earlier, we have $n = 12$

$$\frac{F_M}{n} = \frac{249.38 \text{ N}}{12} = 20.781 \text{ N}$$

Next, we compare the force value in the figure 9.1 and we choose the closest value to it, which is 23.3 N at 90% vacuum level. Corresponding to this we have a vacuum cup with the diameter of 18 mm. By applying a safety factor of 2, it is recommended to select a suction cup with the diameter of 40mm with a theoretical lifting force of 259 N.

Cup		Vacuum Level								
		3 in hg	6 in hg	9 in hg	12 in hg	15 in hg	18 in hg	21 in hg	24 in hg	27 in hg
Diameter	Area	10.2 kPa	20.3 kPa	30.5 kPa	40.6 Kpa	50.8 kPa	61 kPa	71.1 kPa	81.3 kPa	91.4 kPa
mm	cm ²	10%	20%	30%	40%	50%	60%	70%	80%	90%
1	0.01	0.01	0.02	0.02	0.03	0.04	0.05	0.06	0.07	0.07
2	0.03	0.03	0.06	0.10	0.13	0.16	0.19	0.22	0.25	0.28
3.5	0.10	0.10	0.20	0.29	0.39	0.49	0.59	0.69	0.78	0.88
5	0.20	0.20	0.40	0.60	0.80	1.00	1.20	1.40	1.60	1.80
6	0.28	0.29	0.58	0.87	1.20	1.40	1.70	2.00	2.30	2.60
7	0.39	0.39	0.78	1.18	1.60	2.00	2.40	2.70	3.10	3.50
8	0.50	0.52	1.02	1.54	2.00	2.60	3.10	3.60	4.10	4.60
10	0.79	0.80	1.60	2.40	3.20	4.00	4.80	5.60	6.40	7.20
15	1.77	1.80	3.60	5.41	7.20	9.00	10.8	12.6	14.4	16.2
18	2.55	2.60	5.20	7.79	10.4	13.0	15.6	18.1	20.8	23.3
20	3.14	3.20	6.40	9.60	12.8	16.0	19.2	22.4	25.6	28.8
25	4.91	5.00	10.0	15.0	20.0	25.0	30.0	35.0	40.0	45.0
30	7.07	7.20	14.4	21.6	28.8	36.0	43.2	50.4	57.6	64.8
35	9.62	9.80	19.6	29.4	39.2	49.0	58.9	68.6	78.5	88.2
40	12.6	12.9	25.6	38.5	51.2	64.0	76.9	89.6	103	115
50	19.6	20.1	40.0	60.1	80.0	100	120	140	160	180
60	28.3	28.9	57.6	86.5	115	144	173	202	231	259
75	44.2	45.2	90.0	135	180	225	270	315	360	405
80	50.3	51.4	102	154	205	256	308	359	410	461
90	63.6	65.1	130	195	259	324	389	454	519	583
95	70.9	72.5	144	217	289	361	434	506	578	650
110	95.0	97.2	194	291	387	484	581	678	775	871
120	113.1	116	230	346	461	576	692	807	922	1037
150	176.7	181	360	541	720	900	1081	1260	1441	1620
200	314.2	321	640	961	1279	1601	1922	2241	2562	2880

Figure 9.1. Table of theoretical lifting force per cup with identified values

10. Calculating the static frictional force

In order to find the force of static friction, first we determine the normal force applied by the suction cups. Since we are using 12 suction cups, the total suction force is as it follows:

$$F_{total} = F_M * 12 = 249.38N * 12 = 2992.56N$$

Taking into consideration that the coefficient of static friction between the cleaning surface and glass is around $\mu_{f,c/g} = 0.4$ we derive the value of cleaning static frictional force with the following relation:

$$F_{f,c/g} = \mu_f * F_{total} = 0.4 * 2992.56N = 1197.024N$$

In concordance with the figure 10.1, as long as the force of static friction from the wheel rubber on the glass is greater than the static friction of the cleaning material on the glass, the robot will move.

Taking into consideration the static coefficient of friction of the rubber on glass of $\mu_{f,r/g} = 2$ we have:

$$F_{f,r/g} = \mu_{f,r/g} * F_{total} = 2 * 2992.56N = 5985.12N$$

$$(F_{f,r/g} = 5985.12N) > (F_{f,c/g} = 2 * 1197.024N = 2394.048N)$$

Since the condition is met, the robot can move sideways provided the motors supply the required torque. Next we have to test what will happen when the robot tries to move vertically up the window. Since we assumed a total mass of the robot as being $M = 10kg$ we have a total weight force of $F_g = 10 \sin\theta$ where θ is the angle of incline of the robot

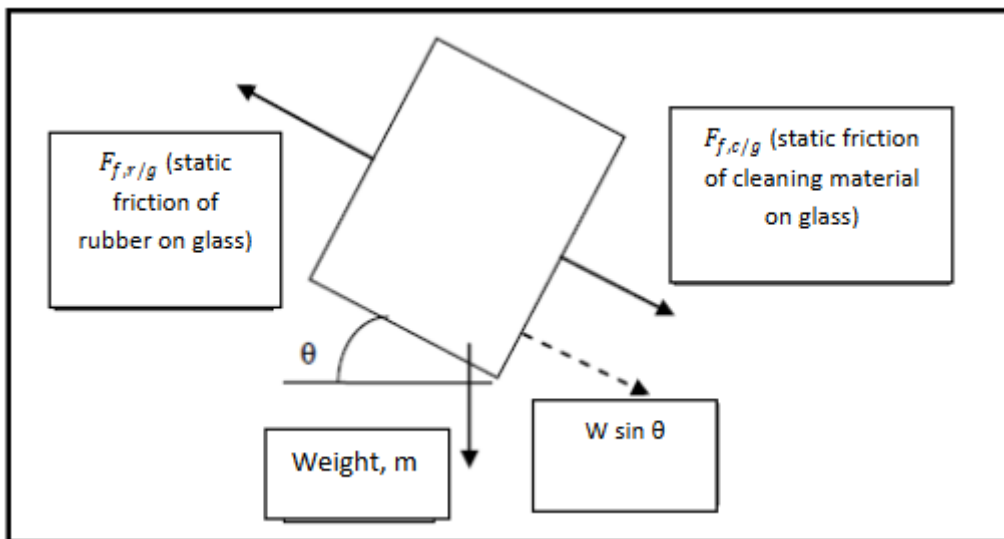


Figure 10.1. Force of static friction of the robot

We have to consider that the force of static friction from the suction cup on the glass must overcome the static friction force of the cleaning surface on the glass as well as the component of weight along that direction.

Since the material is sliding on the glass, normally a kinetic friction force would be used, but the static friction force will be greater and may have to be overcome if the robot stops in this position.

With a static coefficient of friction of rubber on glass of $\mu_{f,r/g} = 2$ we have:

$$F_{f,r/g} = \mu_{f,r/g} * F_{total} = 2 * 2992.56N = 5985.12N$$

$$F_{f,r/g} > \mu_{f,c/g} + F_g \sin\theta$$

$$5985.12N \geq [1197.024N + (10\sin\theta) * 9.81] = (1197.024N + 98.1\sin\theta)N$$

Since this condition is met regardless of the angle to the horizontal, the robot will also move vertically up the window.

For a better understanding of the topics discussed before, in figure 10.2 we have the forces acting on the robot

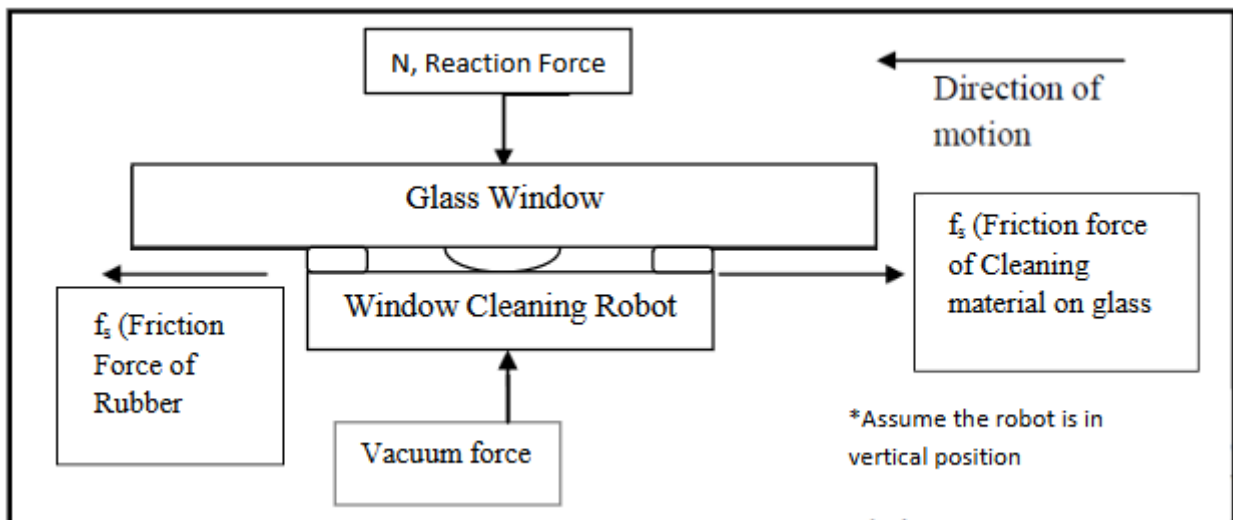


Figure 10.2. Forces acting on the robot

11. Geometry and required pass calculations

The number of passes the device needs to make across the window can be minimized by relating it to the height of the device. For the purposes of this calculation, a single pass is defined as one horizontal movement and one diagonal movement.

Let us consider that the average window size of high building usually is 130cm x 200cm as shown in the figure 11.1.

Assuming the cleaning surface is the same height as the robot, the relationship between the Device's height (H) and the number of passes required to cover the entire window (N) can be represented in the following relationship while assuming that the robot is in square shape, the height x width of the robot is equal to H^2

$$N = \frac{200cm * 130cm}{H^2} * \frac{1}{E}$$

In the above relationship, the value E is the efficiency factor and represents the percentage of repeated area wiped. For example 100 percent efficiency would correspond to $E = 1$, which in turn corresponds to zero overlap when cleaning in successive horizontal passes, while 50% efficiency would correspond to $E = 1$, which indicates that half of the previously-cleaned area is overlapped in the next pass.

11.1. Window shape estimation

Using another alternative, we can estimate the shape of the window.

Because the robot is not able to capture the overall shape of the window panel at one time, the window shape estimation is done by measuring the relative distance between the robot and the window frame with the use of an 2D laser range scanner along the frame and arranging the points obtained by the scanner according to the pose of the robot.

We can use the Extended Kalman filter in order to estimate the pose of the robot, with the use of the odometry data set of the robot.

This way we can construct the pose graph of the robot and record the position of the points obtained by the 2D scanner.

Once it is detected that the robot reaches the end point of the loop of the robot trajectory, the pose adjustment is carried out to increase the accuracy of the window shape estimation.

The flowchart of the window shape estimation concept is presented in the figure 11.1, as seen on the next page.

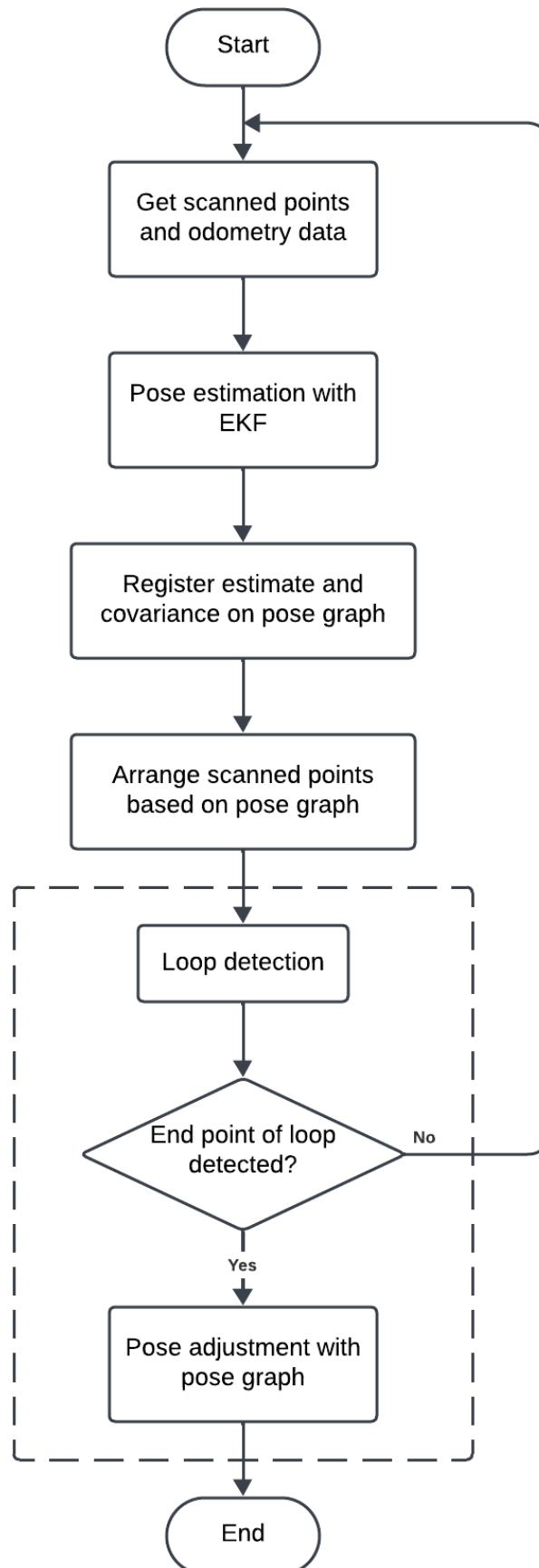


Figure 11.1. Flowchart of the window shape estimation

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2. „Theoretical Holding Force of a Suction Cup” - <https://www.schmalz.com/en/vacuum-knowledge/the-vacuum-system-and-its-components/system-design-calculation-example/theoretical-holding-force-of-a-suction-cup/>
3. „Design and development of a glass facade cleaning robot” - <https://www.sciencedirect.com/science/article/abs/pii/S0094114X2100327X>

4TH SEMESTER

ADJUSTMENTS TO THE WORK PLAN

1. ADJUSTED OBJECTIVES AND ACTIVITIES FOR SEMESTER 4 IN RELATION TO THE ORIGINAL PLAN

The 4th semester will be reserved for improving the existing solutions for the facade cleaning robot, including a partial overhaul of the 3D model. Outside of that, the required recalculations will be made if certain properties of the components have been altered greatly.

4TH SEMESTER RESULTS

12. Adjustments made regarding the 3D model

In regards of the 3D model of the facade cleaning robot, that was done in the second semester, i decided to rebuild it. More precisely i've done a series of modifications to certain components so that the functionality of the robot could be justified while also replacing obsolete features that were not discussed in the project simply because the model realised in the second semester was merely a placeholder.

Outside of that, the control cluster has been integrated in a protective case for protection and utility.

12.1. Locomotion mechanism adjustments

The first ajdusment made was with regards to the locomotion mechanism. It works on the same principle of caterpillar tracks but now the mechanical structure is more defined as seen in the figure 12.2.

The catterpillar tracks are moved through the use of a gear transmission that includes two pairs of gear mechanisms actioned by two actuators, DC motors, as seen in the image 12.3.

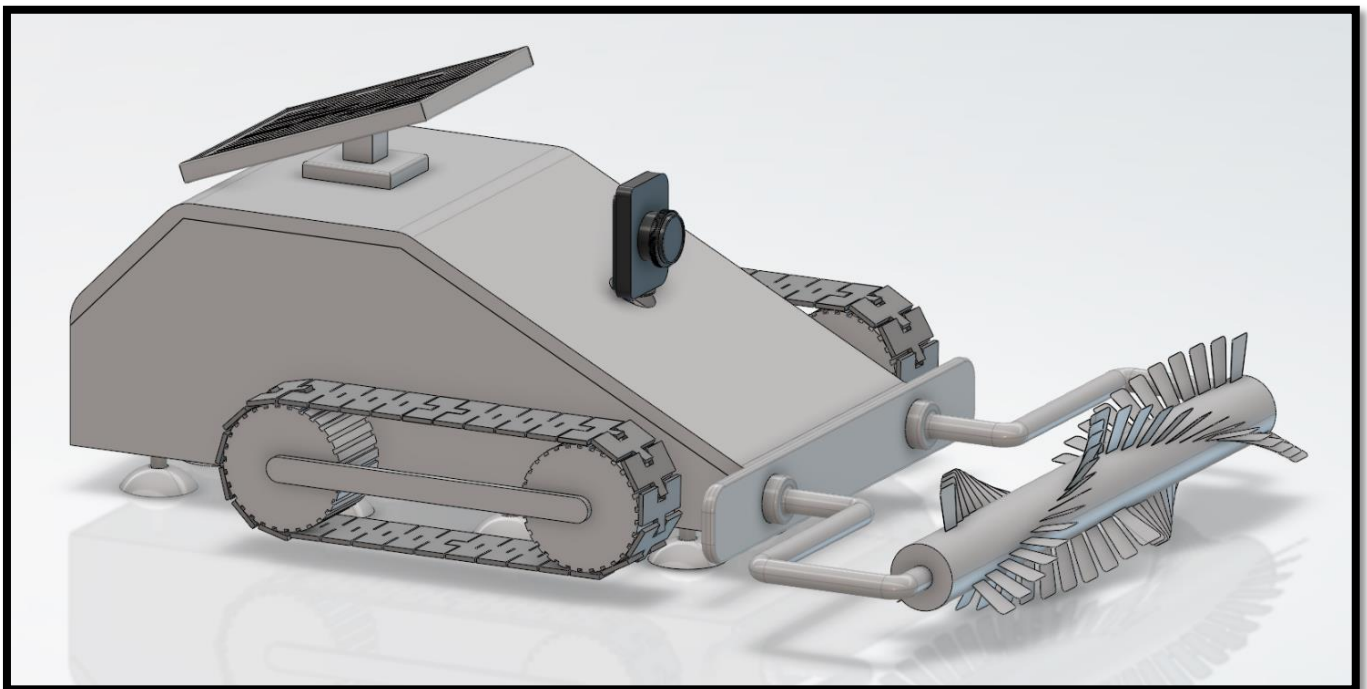


Figure 12.1. The new model of the Glass Facade Cleaning Robot

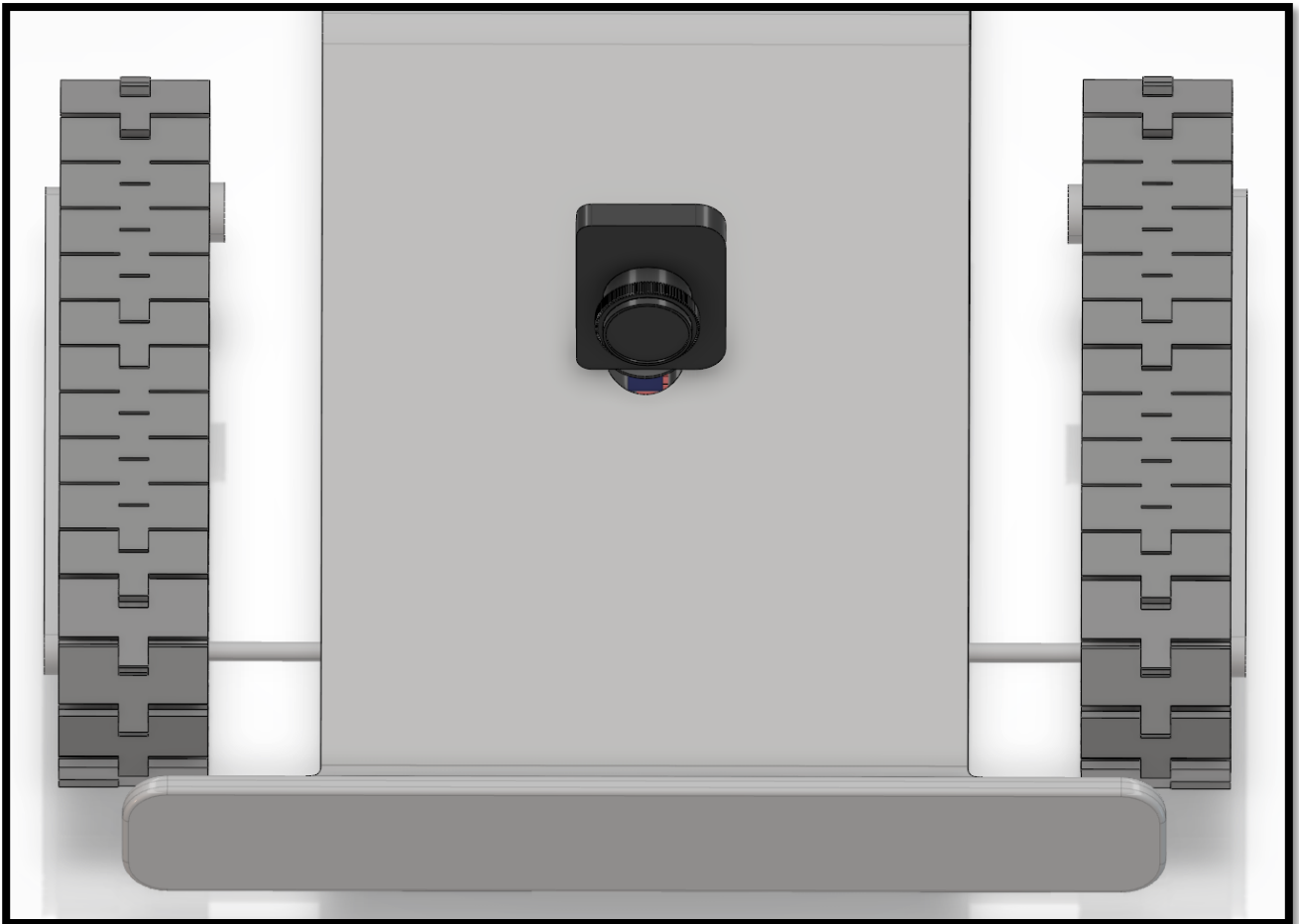


Figure 12.2. The caterpillar mechanism

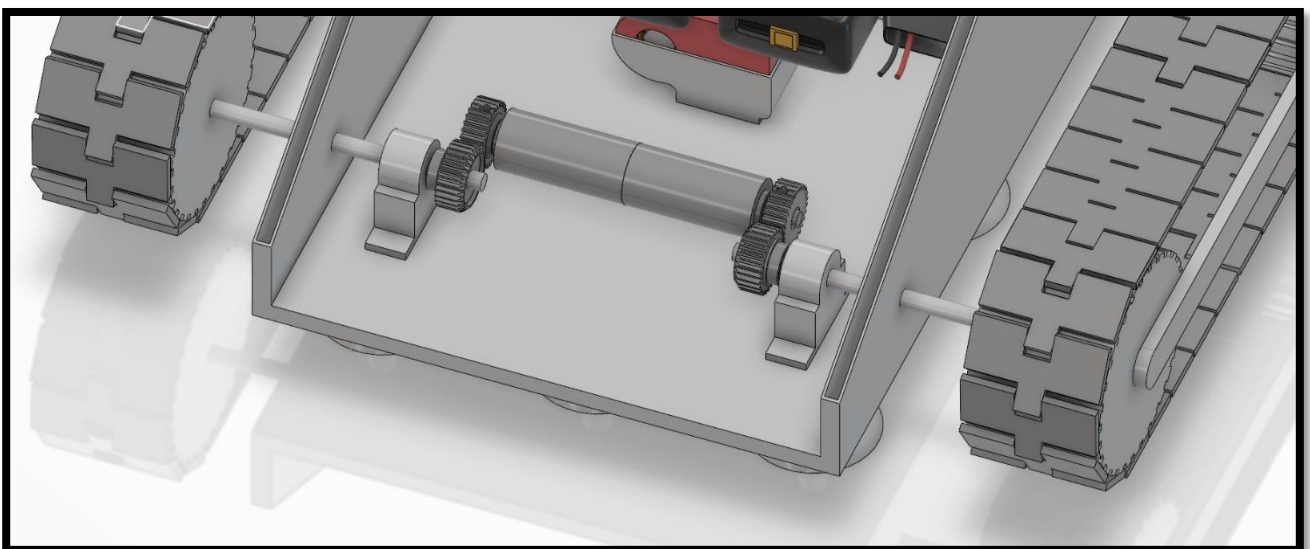


Figure 12.3. The gear transmission mechanism of the caterpillar tracks

12.2. Robot structure adjustments

The adjustments made regarding the robot structure are made, as i mentioned above, with regards to the control sistem and the electrical components that have been placed in a compact, encased structure, as it can be seen in the figure 12.4.

Structural improvements have been made with regards to the driver modules of the motors that are now attached properly as seen in figure 12.5. and also to the entire structure that consists of battery packs, component boards and the raspberry pi which is now properly seated in a plastic case as seen in the figure 12.6.

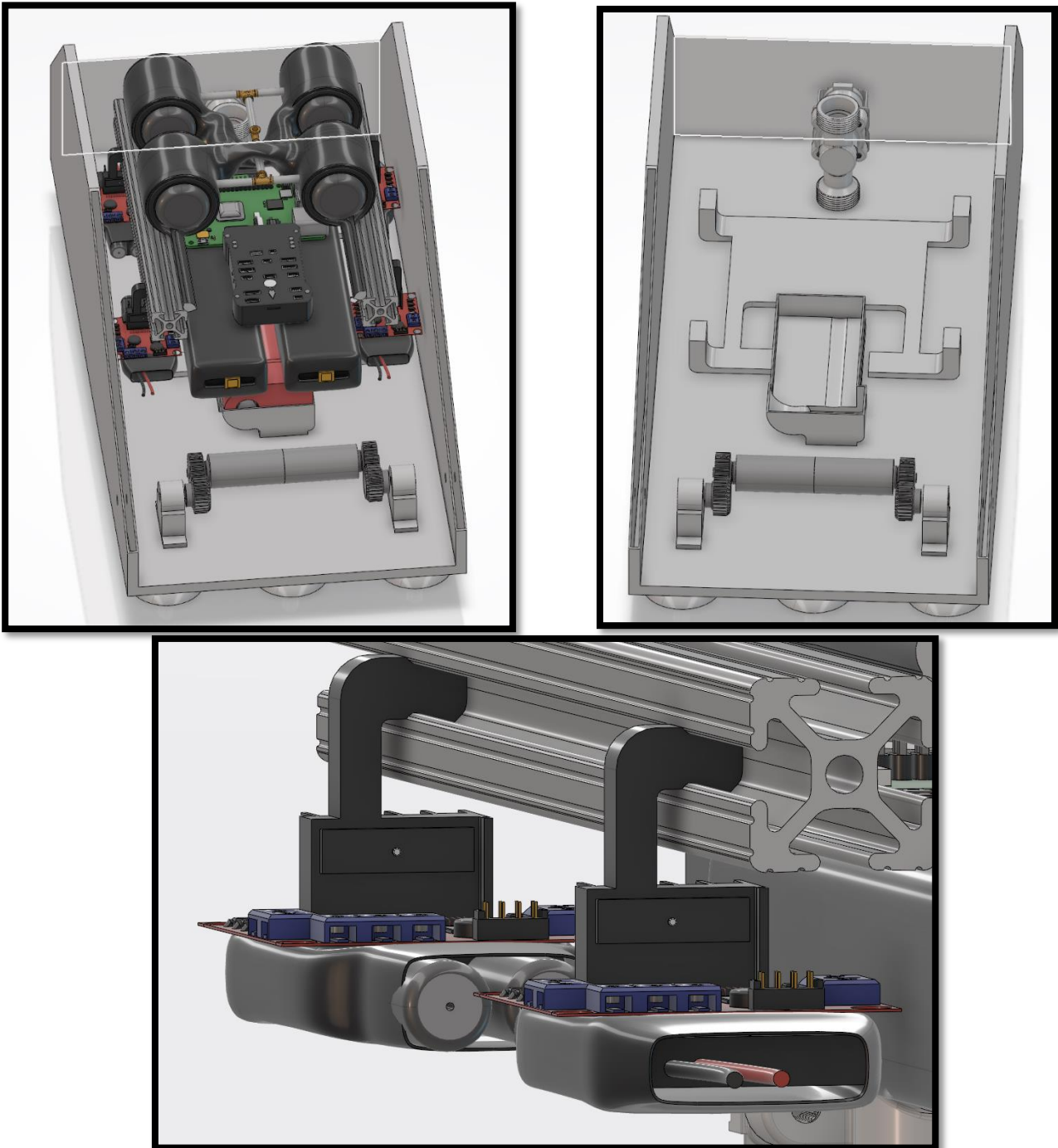


Figure 12.4, 12.5, 12.6. Different views of the components

12.3. Adhesion solution adjustments

Further improvements have been brought to the adhesion solution of the facade cleaning robot, more precisely we integrated the solenoid valve in the mechanism, meaning that now we can regulate the vacuum generated in the suction cups from the control side.

The suction cups are connected in between them through a tube where the vacuum is distributed and controlled through the use of the solenoid valve as seen in figure 12.7.

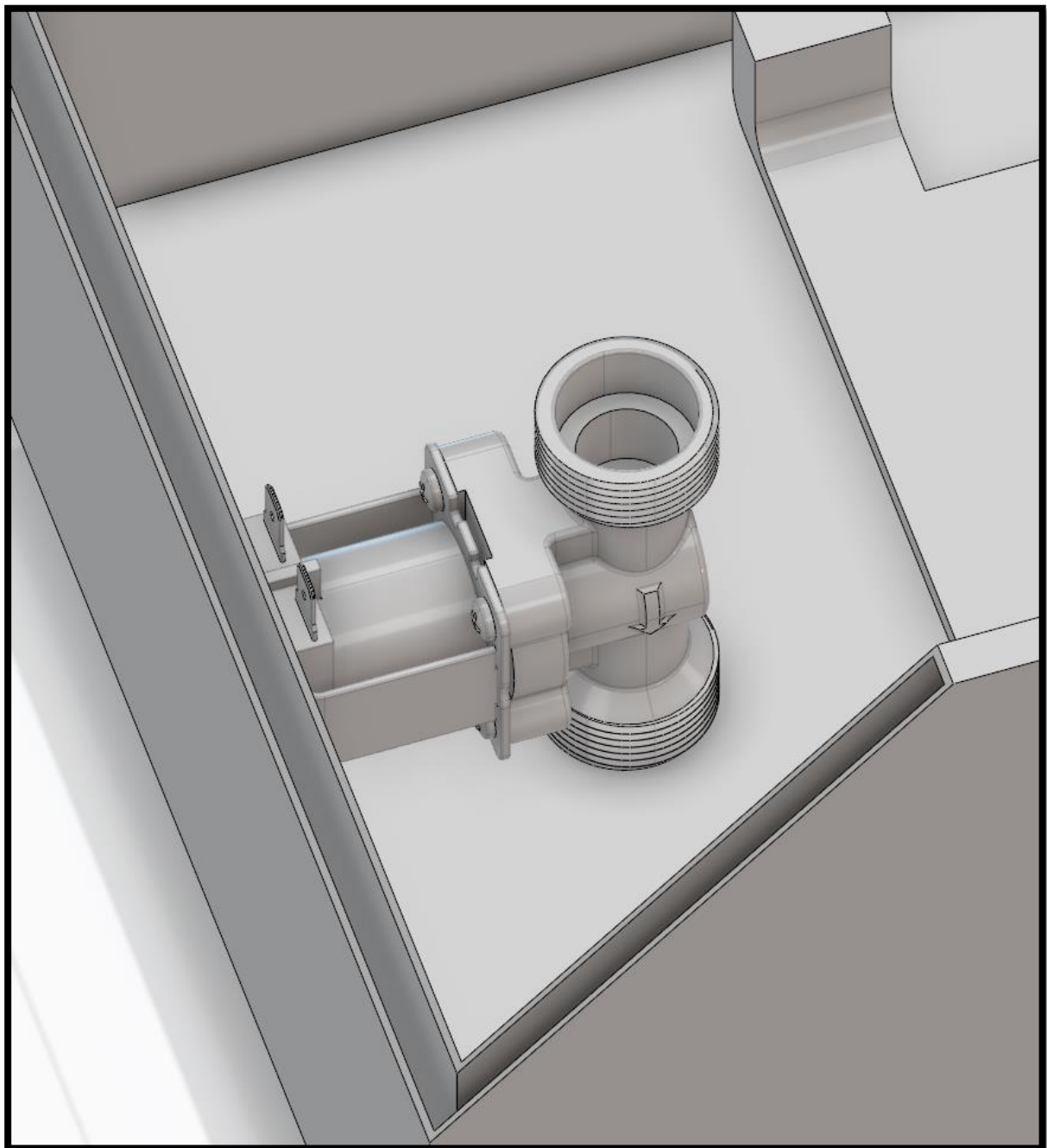


Figure 12.7. The solenoid valve

12.4. Cleaning solution adjustments

Regarding the cleaning solution, an important aspect of it is represented by the reservoir that where the water mixes with the cleaner after which it traverses through an internal piping system that ends with a nozzle that distributes it evenly on the bristle roller, as seen in the images 12.8, 12.9 and 12.10.

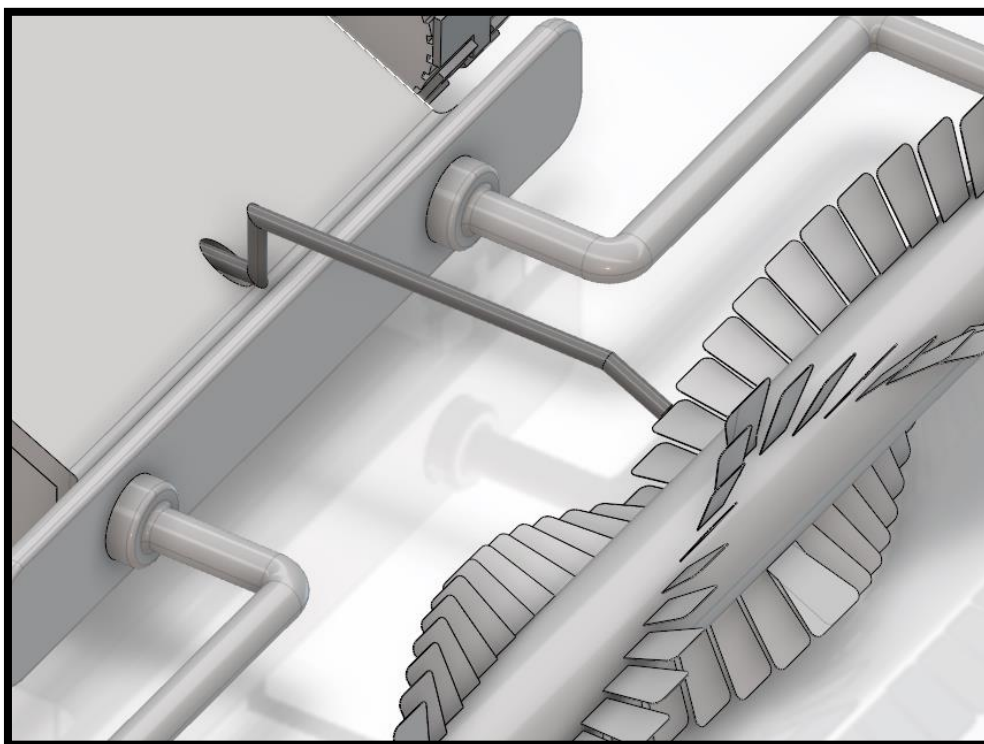
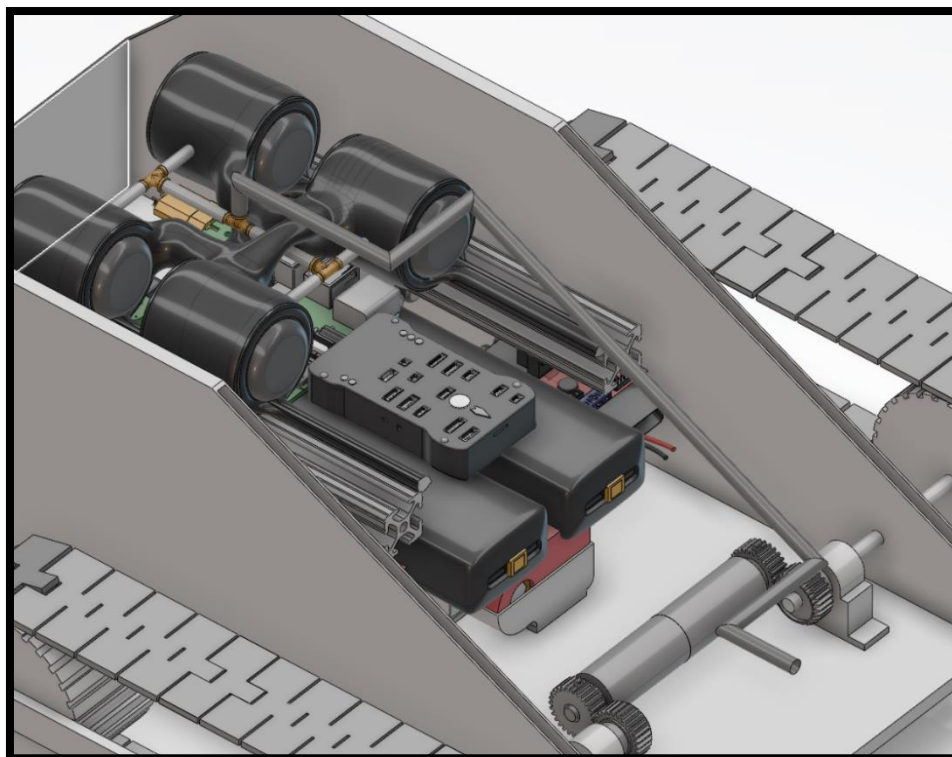


Figure 12.8, 12.9. The reservoir and piping system.

Outside of the adjustments mentioned above, we modified the position of the Lidar camera and of the solar panel on the rooftop of the robot.

The bristle roller is mounted on the front of the robot as seen in the figure 12.9. on the previous page.

By making these adjustments we reduced the space occupied by the robot on the glass pannel while also reducing the total weight and improving the already existing qualities of itself.



Figure 12.10. Water nozzle concept

13. Cost component table of the robot

Now that we know exactly what the project consists of, we can evaluate and create a cost component table that includes all the components costs of it.

We have the table as it follows:

Component	Quantity	Unit Price (USD)	Total Price (USD)	Notes
Raspberry Pi (Model 4B - 4GB RAM)	1	\$55.00	\$55.00	Core controller
FS-iA10B Signal Radio Receiver	1	\$18.00	\$18.00	Used for RC communication
V6 Battery Pack (Li-ion, 12V/3Ah)	2	\$25.00	\$50.00	Power supply for motors/electronics
L298N Motor Driver Module	4	\$5.00	\$20.00	Dual H-Bridge controller for motors
Intel RealSense Lidar Camera (L515)	1	\$350.00	\$350.00	3D mapping, obstacle detection
V4 Linear Actuator (12V, 100mm)	2	\$45.00	\$90.00	For mechanical movement and positioning
Mini Solar Panel (200x150mm, ~10W)	1	\$15.00	\$15.00	Supplemental power, possibly for trickle charging
Solenoid Valve (12V, Normally Closed)	1	\$12.00	\$12.00	For fluid/air control, possibly suction or cleaning fluid release
Vacuum Suction Cups (50mm)	12	\$3.00	\$36.00	For gripping glass surfaces

Assembly & Labor Cost

- Electronics Assembly & Wiring: 4 hours @ \$40/hour = \$160
- Mechanical Assembly (mounting actuators, suction cups, solar panel): 3 hours @ \$40/hour = \$120
- Programming & Testing: 6 hours @ \$50/hour (higher rate for software skills) = \$300

Total Assembly & Labor = \$160 + \$120 + \$300 = \$580

Enclosure / Frame / Mounting Hardware

- Custom aluminum frame, mounts, fasteners, protective casing: \$120

Safety / Redundancy Components

- Emergency stop switch, fuses, protective covers, backup sensors: \$75

In the end we have a grand total cost estimate of **\$1421**

Cost Category	Amount (USD)
Components	\$646
Assembly & Labor	\$580
Enclosure / Frame / Hardware	\$120
Safety / Redundancy	\$75
Total	\$1,421